

# Continuous renal replacement procedures in the intensive care unit

Understanding and differentiated application

Daniel Heise



Springer

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This book is a translation of the original German edition „Kontinuierliche Nierenersatzverfahren auf der Intensivstation“ by Heise, Daniel, published by Springer-Verlag GmbH, DE in 2021. The translation was done with the help of artificial intelligence (machine translation by the service DeepL.com). A subsequent human revision was done primarily in terms of content, so that the book will read stylistically differently from a conventional translation. Springer Nature works continuously to further the development of tools for the production of books and on the related technologies to support the authors.

ISBN 978-3-662-65309-8      ISBN 978-3-662-65310-4 (eBook)  
<https://doi.org/10.1007/978-3-662-65310-4>

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Cover design: deblik, Berlin

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## Preface

Continuous renal replacement procedures have been an integral part of modern intensive care medicine for decades, and currently, it is hard to imagine treating critically ill patients without them. Since the first “blood washes” at the beginning of the twentieth century, knowledge in the field of renal replacement therapy has increased considerably. The practical application of the devices has become easier over the course of time due to constant technical development; today, almost every intensive care unit has the possibility of effectively treating patients with acute kidney failure. The widespread use of renal replacement therapy devices means that renal replacement procedures are also frequently used by non-nephrologists in intensive care units.

Previously valid medical knowledge is often superseded by newer, evidence-based findings from clinical research, and thus the medical-therapeutic aspects of renal replacement therapy (e.g., indications, therapy duration and procedures) are also subject to constant change. However, the underlying physical principles of renal replacement therapy, such as diffusion and convection in the countercurrent process, naturally remain unchanged in their basic features despite all technical advancements and newly gained medical knowledge.

The aim of this book is primarily to convey knowledge about the basic principles and modes of operation of modern renal replacement procedures; clinically oriented information, on the other hand, is deliberately kept to a minimum against the background of its constant change. However, a solid understanding of

the different forms of therapy, their modes of operation and their advantages and disadvantages forms the basis for understanding the currently valid clinical recommendations, for reviewing them in individual cases, for modifying them if necessary and for implementing them in optimal therapy concepts in intensive care practice.

Göttingen, Germany

Daniel Heise

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## Acknowledgements

I would like to thank Mr. Michael Hoffmann of MedicalGraphics Cologne (Germany) for the creative and professional implementation of my ideas in the preparation of the illustrations, as well as PD Dr. Markus Frank of the Electron Microscopic Center of the University Hospital Rostock (Germany) for the unbureaucratic preparation of the electron microscopic images used in this book.

I would also like to express my special thanks to my long-term professional companion Dr. Tassilo Kunzmann and PD Dr. Orfeas Liangos, who proofread my manuscript with the critical eye of an experienced nephrologist and provided me with constructive suggestions for improvement at one point or another.

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## About the Author

**Daniel Heise** has been a member of staff at the Department for Anaesthesiology and Intensive Care Medicine at the University Hospital Göttingen, Germany, since 1995, where he works as a senior physician in the field of intensive care medicine.

During his doctoral thesis on various markers of acute renal failure, completed in 1996, a long-term enthusiasm for this fascinating organ developed. Further scientific activities therefore focused on clinical aspects of acute renal failure.

In addition to his passion for emergency and intensive care medicine, teaching is also particularly close to the author's heart. Making supposedly complicated things simple, conveying connections instead of merely enumerating facts—these approaches formed the basis for the idea of writing this book on continuous renal replacement therapy in intensive care medicine.



# Historical Developments

# 1

At the beginning of the twentieth century, the complete loss of kidney function usually led to the death of the patient. With an increasing understanding of the physiological processes in the kidney, the search began for technical ways to remove substances requiring elimination from the blood of patients with lost kidney function. The first kidney replacement procedures in the 1910s were technically very complex and were not able to reduce the mortality of kidney failure in any relevant way. Technical developments made renal replacement procedures easier to handle, which contributed to their increasing use. However, due in part to the limited sterility of the procedures, they were only ever used intermittently (i.e., for a few hours) until the 1960s. It was not until the industrial production of sterile and pyrogen-free hollow fibre membranes that the way was paved for the continuous use of renal replacement procedures, as is common today in intensive care medicine.

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## 1.1 A Short Journey Through Time

As early as the nineteenth century, the British chemist Thomas Graham discovered a phenomenon that still plays a central role in kidney replacement procedures today. He was able to show in experiments that under certain circumstances, dissolved particles

diffuse through membranes from one fluid compartment to another. Graham's systematic investigations also revealed the two conditions that must be met for diffusion to occur between two compartments: first, the concentrations of the substance in question must be different in the two compartments, and second, the size of the diffusing substances must not exceed a certain maximum, which in turn is determined by the molecular properties of the membrane ("pore size"). This selective diffusion of dissolved particles caused by a difference in concentration was termed "dialysis" by Graham as early as 1854, thus coining a central term in modern renal replacement therapy long before the functional principles of the biological kidney were even rudimentarily known.

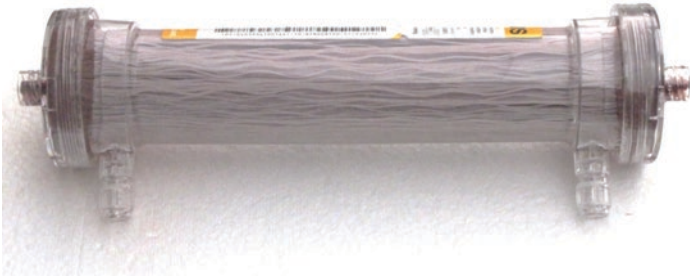
In the early twentieth century, as research provided ever deeper insights into the physiology of the kidney and the functioning of the nephron became better understood, intensive research began into the pathophysiological and pathobiochemical effects of complete failure of excretory renal function. With the realization that in all forms of renal failure, the accumulation of toxic metabolites ultimately leads to the death of the patient, the search began for ways to rid the blood of substances that would be eliminated by the kidney under physiological conditions. At that time, all these experimental procedures were based on the principle of dialysis discovered by Graham: blood and a suitable fluid ("dialysate") flowed past each other in countercurrent, separated by a membrane; certain substances diffused from the blood into the dialysate and were thus eliminated from the blood.

The first practical application of such a method was achieved by the US physician and pharmacologist Johann Jacob Abel in 1912 with the extracorporeal "vividiffusion" system he developed. Abel only used this system for demonstration purposes on experimental animals, which he had previously injected intravenously with various tracer substances. However, after connection to the vividiffusion system and an appropriate treatment period, tracer substances could be detected to a large extent in the dialysate, thus proving the fundamental possibility of selectively removing certain substances from the blood.

The procedure did not come into clinical use until more than 10 years later, when the physician Georg Haas performed the first 11 “blood washes” on patients at the University Hospital in Giessen (Germany) between 1924 and 1928. His apparatus was also based on allowing blood to be washed around a dialysate fluid separately through a semipermeable membrane, whereby dissolved substances passed from the blood into the dialysate. However, the procedure was not widely used at that time due to the very high technical complexity and the continued high lethality of patients with anuric kidney failure. It was only after technical developments by Willem Johan Kolff in the Netherlands (“drum kidney”) and Nils Alwall in Sweden (ultrafiltration by negative pressure) that renal replacement therapy became increasingly widespread in the clinical treatment of patients from the 1950s onward.

Despite extensive technical advances, cellophane tubes from sausage production (!) were initially still used as dialysis membranes, through which the patient’s blood was passed and brought into contact with the surrounding dialysate. Due to the very limited sterility and direct blood contact, “blood washes” represented a relevant risk of infection, so they were only used for a few hours in patients with acute or terminal kidney failure but at a high intensity. Treatment was only resumed when the patient’s water and electrolyte status required it (usually after a therapy-free interval of several days). This intermittent form of therapy, with its abrupt, intensive interventions in the fluid and electrolyte balance, often posed a challenge to the physiological mechanisms of the patients’ circulatory regulation.

It was not until the end of the 1960s that the basis for the continuous use of renal replacement procedures was created with the industrial production of sterile haemofilters. The core components of these haemofilters were several thousand microcapillaries (also known as hollow fibre membranes) made of cellulose or synthetic polymers connected in parallel and surrounded by a fluid-tight cartridge. Blood flowed through these capillaries on the one hand, and dialysate washed around them on the other, which was located in the interior of the cartridge. Due to the large number of capil-



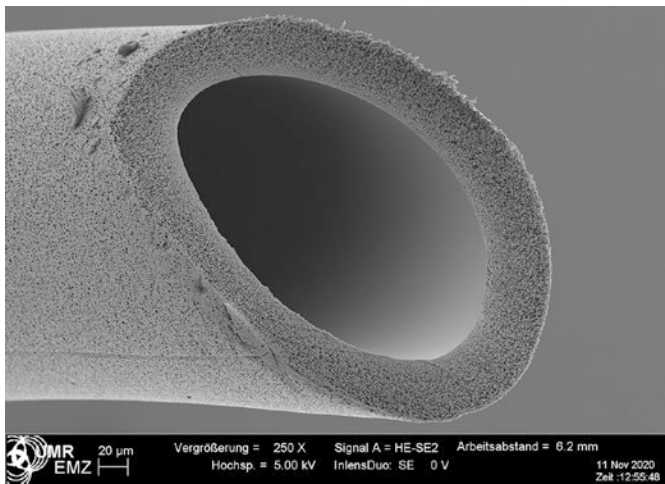
**Fig. 1.1** Modern haemofilter with hollow fibre membranes. The side ports direct blood flow through the microcapillaries, while the bottom ports provide access to the extracapillary space of the filter cartridge. (Own representation)

larities, the contact surface between blood and dialysate reached a size of approximately  $1 \text{ m}^2$  despite the compact design of the product and the small diameter of the capillaries (approx.  $200 \text{ }\mu\text{m}$ ). This design has remained virtually unchanged to this day; Fig. 1.1 shows a modern haemofilter with two lateral connections for blood flow through the capillaries and two connections for the extracapillary dialysate compartment of the cartridge.

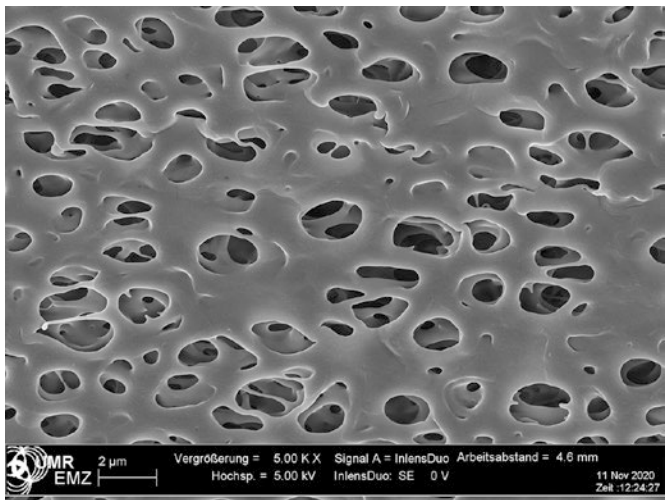
The porous character of the capillaries can be seen in the electron micrograph of a microcapillary (Fig. 1.2). At a higher magnification, the fine structure of the capillary wall can be seen vividly in the top view of the capillary surface (Fig. 1.3).

The term “sieve coefficient” was introduced to describe the filtration properties of capillaries. The sieve coefficient is the ratio of the concentration of a substance outside the capillary to its concentration inside the capillary. If a substance is freely filtered, the concentrations inside and outside the capillary are equal, and the sieve coefficient in this case is 1.0. If, on the other hand, a substance is not filtered at all, its concentration outside the capillary is 0, and the sieve coefficient is therefore also 0.

- ▶ Sieve coefficient = 1.0 : Free filtration of the substance
- ▶ Sieve coefficient = 0.0 : No filtration of the substance



**Fig. 1.2** Electron micrograph of a hollow fibre membrane of a haemofilter. The magnification of 250x reveals the trabecular structure of the capillary wall and the micropores on the surface. (Source: Electron Microscopy Center of the University Hospital Rostock, Germany)

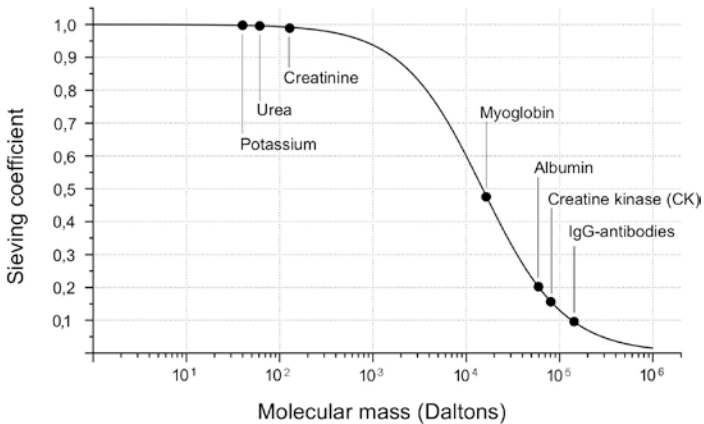


**Fig. 1.3** Top view of the capillary surface. At a magnification of 5000x, a fine structure and the pores of the membrane can be seen very well. (Source: Electron Microscopy Center of the University Hospital Rostock, Germany)



If the sieve coefficients of dissolved substances are plotted in a diagram as a function of their molecular size, characteristic curves are obtained that characterize the filtration behaviour of the respective filtration membrane (see Fig. 1.4). Small molecules such as creatinine and urea are filtered almost completely, whereas large molecules such as albumin or antibodies can only pass through the pores to a small extent (less than 20% and 10%, respectively).

The availability of high-quality, sterile and pyrogen-free hollow fibre membranes represented a significant milestone in the development of renal replacement therapy because it made the continuous use of renal replacement procedures over several days possible for the first time.



**Fig. 1.4** Characteristic curve of a modern haemofilter. Up to a molecular size of  $10^3$  Da, substances are filtered almost freely, and molecules with a size of approximately 150 kDa (e.g., IgG antibodies) can pass through the capillary membrane only to approximately 10%. (Own representation)

# Proven to the Present Day: Continuous Haemofiltration

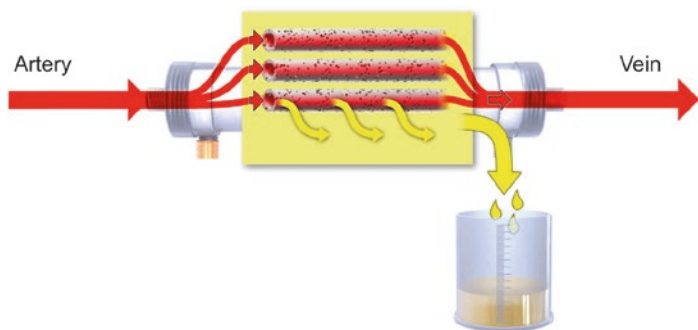
The first continuous renal replacement therapy to be used was arteriovenous haemofiltration in 1977. In addition to its continuous use, the concept of haemofiltration (i.e., the removal of solutes through an outflow of fluid) was also an innovation at the time. The continuous use of renal replacement procedures proved so beneficial after a short time that the originally very simple arteriovenous procedure was refined by a series of technical advancements. In this chapter, the function and significance of all components of modern continuous veno-venous haemofiltration are explained in detail.

---

## 2.1 First Continuous Renal Replacement Therapy

The first continuous renal replacement therapy, i.e., one that lasted several days was carried out in 1977 by the nephrologist Peter Kramer in Göttingen, Germany (Kramer et al. 1977). Figure 2.1 shows the basic structure of the procedure, which featured another innovation in addition to continuous application:

Instead of allowing the blood to be flushed by a dialysate fluid, the removal of urinary substances took place exclusively by filtering plasma fluid and the substances dissolved in it (“solute”) through the pores of the hollow fibre membranes.



**Fig. 2.1** Schematic structure of the first continuous arteriovenous haemofiltration. Solutes are removed by the outflow of plasma fluid, and blood flow and filtration occur passively due to the arterial mean pressure. (Image creation: MedicalGraphics, Cologne, Germany)

- Solutes = dissolved substances in the blood, e.g., electrolytes, creatinine, urea, etc.

The extracapillary space was therefore not filled with dialysate fluid (as was the case with the diffusive dialysis methods practised up to that point), but rather it was rapidly filled with pressed ultrafiltrate, which was able to drain freely.

The connection of the filter was arteriovenous, and the driving force for the blood flow through the filter capillaries was thus the difference between arterial mean pressure and central venous pressure (CVP). This reduced the technical complexity of the procedure to a minimum, as no active pumps were required for blood flow.

However, this simple arteriovenous procedure was associated with several disadvantages due to the system:

- The arteriovenously perfused capillary filter represented an “accessory organ” from a haemodynamic point of view. Therefore, the heart had to be able to effectively increase cardiac output to generate effective blood flow through the filter in addition to sufficient perfusion of the biological organs.

- The amount of blood flow through the filter capillaries was determined exclusively by the difference between arterial and central venous pressure and the flow resistance of the filter capillaries. Because these variables can vary within wide limits, the amount of blood flowing through the filter was also highly variable and could only be influenced to a very limited extent.
- The amount of fluid filtered off per unit time could not be influenced either, as it depended essentially on the hydrostatic pressure inside the filter capillaries, which in turn correlated closely with arterial blood pressure. Although some autoregulation was observed in the first applications of this procedure in 1977, as the amount of plasma fluid filtered off increased, the patient's mean arterial pressure decreased, which in turn led to a decrease in filtrate production. However, despite this gross autoregulation, the extent of fluid removal could only be influenced and controlled to a very limited extent.

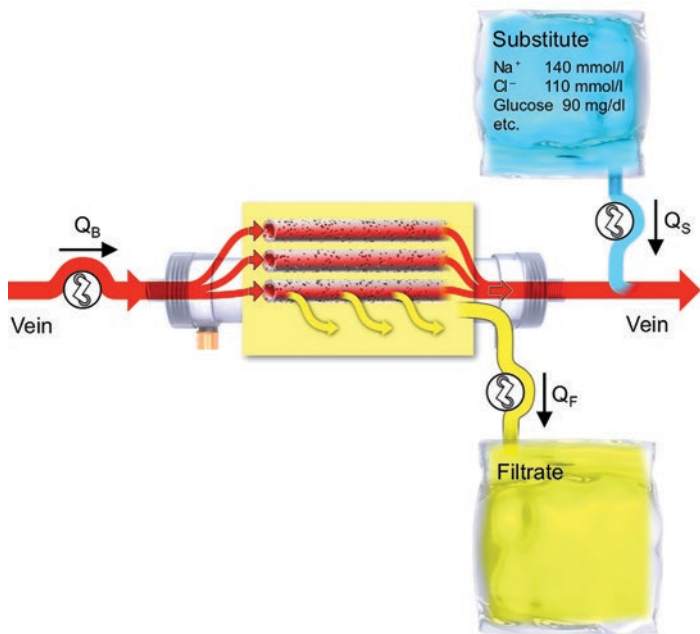
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## 2.2 Technical Developments

However, because the new concept of continuous renal replacement therapy proved to be advantageous compared to therapies that had only been performed intermittently up to that time, technical modifications were developed in the following period to overcome the described disadvantages of the simple arteriovenous system. These technical advancements, which are illustrated in Fig. 2.2, consisted of three components:

### 2.2.1 Blood Pump

By installing a roller pump in the blood supply line, the blood flow was no longer passive-pressure dependent but actively generated by the pump. This enabled the veno-venous connection to the patient, as an arterio-venous pressure gradient was no longer required for the perfusion of the filter.



**Fig. 2.2** Schematic structure of continuous veno-venous haemofiltration (CVVH). In addition to the blood flow ( $Q_B$ ), the flow rates of the filtrate ( $Q_F$ ) and substitute ( $Q_S$ ) are also controlled by roller pumps. (Image creation: MedicalGraphics, Cologne, Germany)

Although modern renal replacement procedures are virtually exclusively veno-venous, the designations of the blood supply and return lines are still a relic from the days of historical arteriovenous procedures: The blood flows to the device via the “arterial” connection and back to the patient via the “venous” connection.

The veno-venous connection not only eliminated the need for large-lumen cannulation of arterial vessels but also removed the negative effects of the arterio-venous procedure on haemodynamics (no increase in cardiac output required). Another advantage of the active blood pump was that the level of blood flow could be precisely adjusted and kept constant via the delivery rate of the blood pump; this defined blood flow is referred to as  $Q_B$ .

### 2.2.2 Filtrate Pump

The outflow of the ultrafiltrate was also no longer passive and dependent on the hydrostatic pressure within the capillaries but was regulated by a second roller pump, the filtrate pump. This made it possible to control exactly how much ultrafiltrate flows out per unit of time; this defined filtrate flow is referred to as  $Q_F$ .

Because the ultrafiltrate is also pressed out of the capillaries in veno-venous operation with a high hydrostatic pressure (generated by the blood pump), the function of the filtrate pump may well be to *limit* the effective filtrate flow if the spontaneous filtrate outflow due to the hydrostatic pressure would be higher than the therapeutically desired filtrate flow. This flow limitation by the filtrate pump is comparable to the “engine brake” of a car when the engine is not used to accelerate the vehicle when driving downhill but to brake it.

However, if high filtrate flows are desired, the filtrate pump may well increase the flow by generating an appropriate vacuum in the extracapillary compartment of the filter cartridge, thereby actively drawing fluid from the hollow fibre membranes.

### 2.2.3 Substitution Solution

In addition to the two active pumps for blood and filtrate flow, a third further development became established, which is still an integral part of modern haemofiltration procedures. After the blood has passed through the haemofilter and the desired amount of ultrafiltrate has been filtered off, a special substitution solution is added to the blood before it is retransfused to the patient by returning venous access. This inflow is also generated by a separate pump and is referred to as the substitution flow  $Q_s$ .

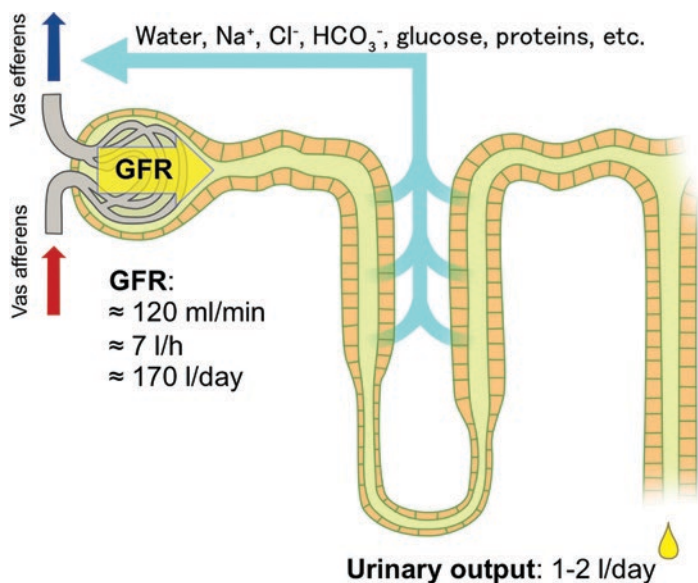
Most patients are volume-expanded (i.e., “overhydrated”) at the beginning of renal replacement therapy so that effective fluid removal with a negative balance is attained. This raises the legitimate question of what the point of such a high-dose fluid intake is—in everyday clinical practice, substitution flows in the range

of 2000 ml/h (i.e., approximately 50 l per day!) are common. To understand the sense and necessity of such a high-dose fluid supply, the details of the elimination of urea should be considered in more detail; here, urea is to be seen as an example for almost all other solutes.

Protein metabolism in humans causes an accumulation of approximately 15 g urea per day, which must be completely eliminated via the renal replacement procedure when renal function ceases. The serum concentration of urea, however, is only on the order of approximately 30 mg/dl, i.e., 300 mg/l or 0.3 g/l. Consequently, to eliminate the total daily amount of 15 g urea from the blood by the mechanism of filtration, 50 l of fluid would have to be squeezed out of the bloodstream in 24 h ( $0.3 \text{ g/l} \times 50 \text{ l} = 15 \text{ g}$ ). However, the sole filtration and excretion of blood fluid on the order of 50 l per day is simply not possible due to the limited extracellular volume (which is only approximately 20% of the body weight).

A comparison of the *quantity* of a substance that must be excreted per unit of time and its *concentration* in the blood illustrates the fundamental problem: for effective and complete excretion of most urinary substances, considerably more plasma fluid must be filtered out than the fluid that may be withdrawn from the organism. Nature has solved this problem in the kidney in an elegant way, the principle of which is illustrated in Fig. 2.3.

In the kidney, too, considerably more blood fluid is filtered off per unit of time than fluid should be removed from the body. At a normative glomerular filtration rate (GFR) of 120 ml/min, more than 170 l of plasma fluid are filtered off by the glomeruli per day, effectively eliminating even those solutes that are dissolved only in very low concentrations in the blood. However, to reduce the actual fluid loss to approximately 1–2 l per day, the tubular system recovers almost the same amount of water and nonexcretory substances (electrolytes, glucose, proteins, etc.) from the primary filtrate and returns this reabsorbed fluid to the bloodstream via the veins of the tubular system. This type of “recycling” avoids high losses of water and other important substances that would otherwise inevitably occur with filtration on the order of 150–200 l per day. This concept thus makes the two central requirements com-



**Fig. 2.3** Functional principle of the nephron. The reabsorption of water, electrolytes and other physiological blood components enables a high filtration performance with minimal fluid losses at the same time. (Image creation: MedicalGraphics, Cologne, Germany)

patible, on the one hand to filter out very large quantities of fluid but on the other hand to withdraw only a small amount of fluid from the organism.

Despite all advances in biotechnology, it has not yet been possible to artificially reproduce the reabsorption function of the tubular system. When haemofiltration is used, the filtered plasma fluid with all the solutes it contains can therefore no longer be recycled and must be discarded completely. To imitate the recycling function of the biological kidney, normally reabsorbed substances (water, electrolytes, glucose, etc.) are therefore added to the blood after the filtration process in the form of a substitution solution. From an economic point of view, this procedure is by far not as elegant as the direct recycling of the ultrafiltrate in the kidney. However, the addition of a substitution



solution makes it possible to reconcile high ultrafiltration (and thus effective elimination of low-concentration substances) with much lower fluid removal.

- Substitution solution: Enables high filtration performance without relevant fluid losses.

---

## 2.3 Influence of the Molecule Size on the Filtration Process

Because the pores of the membrane have a certain maximum size, it is obvious that the filterability of substances beyond a certain molecular size decreases with increasing molecular weight. The relationship between molecular size and filterability has already been exemplified in Fig. 1.4, and characteristic curves of this type are available for all clinically used filtration membranes. Although the transition between free filterability (small molecules) and complete lack of filterability (very large molecules) is fluid, a cut-off value has been defined that characterizes dialysis membranes and is ultimately determined by pore size. The cut-off value is the molecular size at which only 10% of a substance is still filtered off through the pores of the membrane. In Fig. 1.4, for example, this value is slightly above  $10^5$  Daltons or 100 kDa.

- Cut-off value of a haemofilter: The molecule size at which 10% of a substance is still filtered off.

One of the most essential properties of filtration is that all molecules with masses significantly below this cut-off value are eliminated equally effectively. Similar to pieces of flotsam in a river, which all move downstream at the same speed regardless of their size, all solutes below a certain molecular size are carried out of the blood by the fluid outflow through the pores to the same extent. Haemofiltration therefore eliminates all solutes up to a certain molecular size equally effectively.

- Haemofiltration: All molecules whose mass is below the cut-off value are eliminated almost equally effectively.

This finding may seem trivial at first glance. However, it is important because other mechanisms for substance elimination will be presented in the further course of this book, in which the molecular size has a decisive influence on the elimination of all dissolved substances across the entire spectrum.

---

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Kramer P, Wigger W, Rieger J, Matthaei D, Scheler F (1977) Arteriovenous haemofiltration: a new and simple method for treatment of over-hydrated patients resistant to diuretics. *Klin Wschr* 55:1121–1122

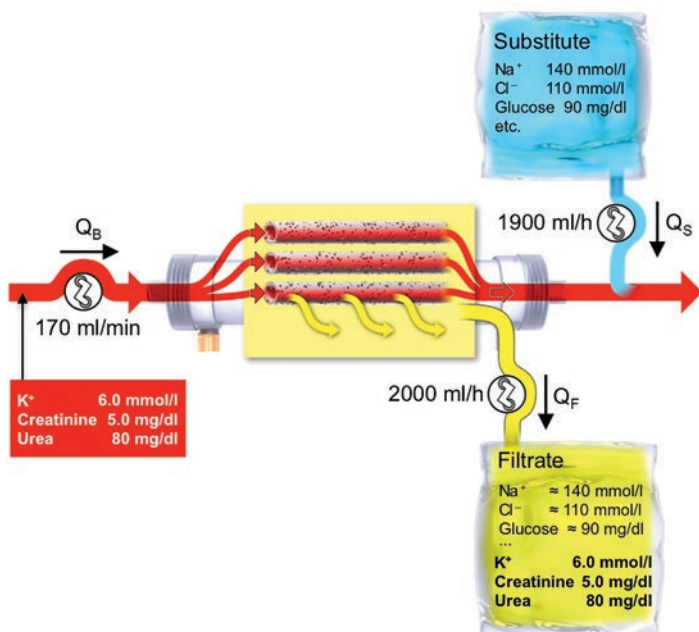
# Parameters of Renal Replacement Therapy

Despite a multitude of different therapy modalities and variants, the therapeutic effect of any renal replacement procedure can always be objectively described with two therapeutic variables: fluid removal and therapeutic dose. The clinical significance of these two important parameters is explained in detail in the following chapter, and the concept of predilution with its advantages and disadvantages is also discussed as a specific variant of haemofiltration.

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## 3.1 Fluid Removal and Therapeutic Dose: Please Do Not Confuse!

Modern renal replacement devices confront their users with a considerable variety of setting possibilities and therapeutic options. At first glance, therefore, it may seem difficult to grasp, assess and, if necessary, optimize the overall effect of a renal replacement procedure. Ultimately, however, all of the parameters that can be set by the user (no matter how many of them there are!) converge in exactly two variables that precisely quantify the therapeutic effect of any renal replacement procedure: fluid removal and therapeutic dose.



**Fig. 3.1** Continuous veno-venous haemofiltration (CVVH) with typical flow rates for blood, filtrate and substitute. Substances requiring elimination have the same concentrations in the filtrate as in the blood plasma. (Image creation: MedicalGraphics, Cologne, Germany)

In continuous haemofiltration (CVVH) discussed thus far, these two quantities are defined by the filtrate flow  $Q_F$  and the substitute flow  $Q_S$ ; accordingly, Fig. 3.1 shows typical values for the filtrate and substitute flow. Based on these flow rates, the two therapeutic quantities of fluid removal and therapeutic dose are explained as follows.

## 3.2 Fluid Removal

The term fluid removal is almost self-explanatory: it refers to the amount of fluid withdrawn from the patient per unit of time by the renal replacement procedure. Fluid removal is therefore the fluid

balance of the renal replacement procedure, whereby the word “removal” by definition contains the sign “−” with regard to the balance. For example, a fluid removal of 100 ml/h is equivalent to a balance of −100 ml/h. Despite many possible variations, fluid removal ultimately always results from the difference between all inflows and outflows of fluids.

- Fluid removal = Outflowing liquids − Inflowing liquids

In haemofiltration, the fluid removal depends on the filtrate flow  $Q_F$  (outflow) and the substitute flow  $Q_S$  (inflow). If 2000 ml of filtrate is withdrawn from the blood per hour, as in Fig. 3.1, and 1900 ml of substitution solution is added at the same time, a net fluid removal of 100 ml/h (corresponding to 2.4 l per day) results. The same fluid removal could be achieved with any other combination of filtration and substitution flow, with a difference of 100 ml/h. The fluid removal of a haemofiltration thus results from the difference between filtrate flow and substitute flow alone.

- Haemofiltration: Fluid removal =  $Q_F - Q_S$

No generally applicable “standard values” can be given for fluid removal; they depend on the individual volume status of the patient and the desired fluid balance. In practice, values between 0 ml/h and 300 ml/h are common, and removal rates of up to 500 ml/h may be indicated in the short term for patients with severe volume expansion.

- Fluid removal: Variable within wide limits.

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### 3.3 Therapeutic Dose

In contrast to fluid removal, which refers to the pure fluid balance, the treatment parameter “therapeutic dose” quantifies the extent to which the blood is freed from substances requiring elimination. The totality of these substances is a highly variable mixture of the most diverse individual substances, whose concentrations in the blood plasma can also vary over very wide ranges. The total amount of all substances removed from the blood with the pressed

filtrate could therefore only be determined by complex quantitative analyses in the ultrafiltrate. However, these are not practicable in routine clinical practice. To be able to easily quantify the elimination of urinary substances, a definition was chosen that is based on the pharmacological concept of clearance: the therapeutic dose describes the volume of plasma fluid (or plasma water) that is completely cleared of urinary substances per unit of time. Its unit is ml/h.

- Therapeutic dose = the amount of plasma fluid that is completely cleared of the substances to be eliminated per unit of time.

It is important to understand here that the concept of a defined plasma volume being completely cleared of its urinary substances is purely virtual. A therapeutic dose of 2000 ml/h does, of course, not free 2000 ml of plasma fluid completely from all solutes in 1 h, while the concentrations of these substances are unchanged in the remaining plasma fluid. However, a therapeutic dose of 2000 ml/h removes the amount of solutes that would be contained in 2000 ml of plasma fluid from the *total plasma fluid* per hour.

Analogous to the body weight-related dosing of drugs, the therapy dose of a renal replacement procedure can also be related to the patient's body weight. The limited significance of absolute values is illustrated by the consideration that a therapy dose of 1500 ml/h corresponds to only 20 ml/kg/h for a body weight of 75 kg but already 30 ml/kg/h for a body weight of 50 kg. For the concrete application of renal replacement procedures, it is therefore advisable to state the therapy dose in ml/kg/h on a weight basis.

In the schematic drawing of a haemofiltration in Fig. 3.1 it is noticeable that a purification process of the blood in the true sense, i.e., the extraction of solutes from the blood, does not take place at all. However, because the filtrate consists of filtered plasma fluid, the concentrations of solutes in the filtrate are just as high as in the blood plasma, as seen in Fig. 3.1. If a defined volume of plasma fluid is filtered off together with all the substances dissolved in it and then replaced by the same volume of

contaminant-free substitution solution, the effect in terms of contaminant elimination is the same as if this volume of plasma fluid had been completely freed of its contaminants. With each unit of volume that flows off as ultrafiltrate and is replaced by substitute, an equal unit of volume of the remaining plasma fluid is virtually freed of its urinary substances. In haemofiltration, the therapeutic dose thus corresponds to the filtrate flow  $Q_F$ .

- Haemofiltration: therapeutic dose = filtrate flow  $Q_F$

In contrast to fluid removal, which can vary within wide limits depending on the patient's current volume status, clearly delineated recommendations on the therapeutic dose level emerged from several independent clinical trials.

A first randomized study from 2000 showed that intensive care patients had a lower survival rate when renal replacement therapy was given at lower therapy doses (Ronco et al. 2000). This is not surprising as substances requiring elimination are also removed from the blood only to a small extent at low therapy doses. Interestingly, however, dose increases beyond 35 ml/kg/h no longer had a positive effect on outcome in this study; the positive effect of higher therapy doses therefore apparently has a saturation characteristic. Two other large-scale randomized trials came to the same conclusion, namely, that increases in therapy dose beyond a certain point no longer improve outcome (VA/NIH Acute Renal Failure Trial Network et al. 2008; RENAL Replacement Therapy Study Investigators et al. 2009). Due to this homogeneous data situation, it is recommended that a value between 20 and 30 ml/kg/h be set as the therapy dose for all continuous renal replacement procedures.

- Therapeutic dose for all continuous renal replacement procedures: 20–30 ml/kg/h.

For a normal-weight patient, this results in a filtration rate of approximately 2000 ml/h or 33 ml/min with CVVH. This corresponds to only approximately one-third to one-quarter of the normal glomerular filtration rate of the kidney. However, a decline in biological kidney function to comparable values

(GFR approximately 30 ml/min) does not usually lead to serious effects on the organism and can normally be compensated for without any problems.

The independence of the variables “therapeutic dose” and “fluid removal” can be illustrated by two examples:

- $Q_F = 4000$  ml/h,  $Q_S = 4000$  ml/h: High therapy dose (4000 ml/h), no fluid removal (0 ml/h)
- $Q_F = 1000$  ml/h,  $Q_S = 600$  ml/h: Low therapeutic dose (1000 ml/h), high fluid removal (400 ml/h)

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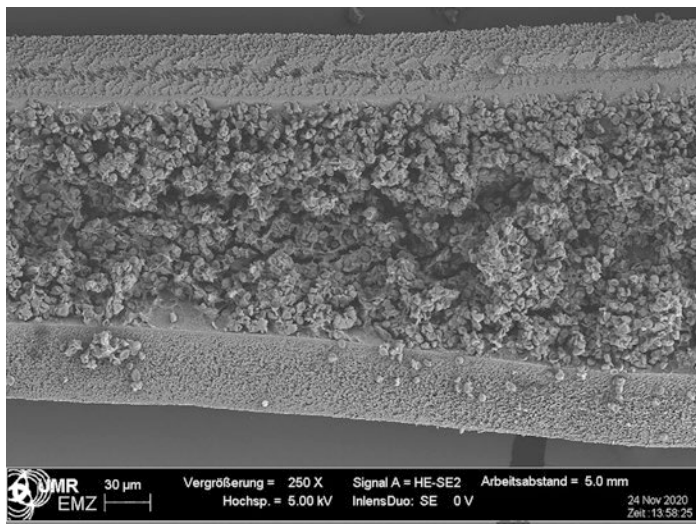
### 3.4 How Should the Blood Flow $Q_B$ Be Set?

While filtrate and substitute flow determine the therapeutic parameters of fluid removal and therapeutic dose in haemofiltration, blood flow  $Q_B$  has no direct influence on these treatment parameters. However, the level of selected blood flow  $Q_B$  also plays an important role, especially in filtration procedures.

Due to the outflow of ultrafiltrate, the haematocrit and the concentration of plasma proteins and coagulation factors increase continuously during passage through filter capillaries. The consequences of this “thickening” of the blood can be seen in Fig. 3.2: The increasing haematocrit leads to an increase in the viscosity of the blood, and in conjunction with the increased concentration of clotting factors, increased clotting processes may occur in the filter capillaries. If a critical part of the filter capillaries is closed by clotting, renal replacement therapy must be discontinued before the end of the maximum permissible treatment period (usually 72 h), i.e., the so-called “filter life” is shortened. However, premature discontinuation of renal replacement therapy not only has therapeutic consequences but also requires human and material resources. The degree of concentration of the blood during its passage through the filter capillaries thus plays an important role against this background.

The greater the amount of fluid filtered off ( $Q_F$ ) in relation to the volume of blood passing through ( $Q_B$ ), the more pronounced the concentration of the blood. The percentage of fluid removed





**Fig. 3.2** Longitudinal section through a microcapillary with clotting phenomena, magnification 250x. (Source: Electron Microscopy Center of the University Hospital Rostock, Germany)

from the blood in the form of filtrate is also called the “filtration fraction”.

$$\blacktriangleright \quad \text{Filtration fraction (in\%)} = \frac{\text{Filtrate flow } Q_F^*}{\text{Blood flow } Q_B} 100$$

To prevent excessive thickening of the blood with the negative effects described, no more than 20% of the volume of the blood in the filter capillaries should be removed by ultrafiltration; the filtration fraction should therefore be a maximum of 20%.

$$\blacktriangleright \quad \text{Filtration fraction: Maximum 20\%}$$

A defined filtrate flow  $Q_F$  (= therapy dose!) therefore always requires a blood flow at least five times as high; for a filtrate flow  $Q_F$  of 2000 ml/h, the blood flow  $Q_B$  must therefore be at least 10,000 ml/h or approx. 165 ml/min (note: in contrast to filtrate and substitute flows, the blood flow of a renal substitution procedure is given in millilitres per *minute*!).

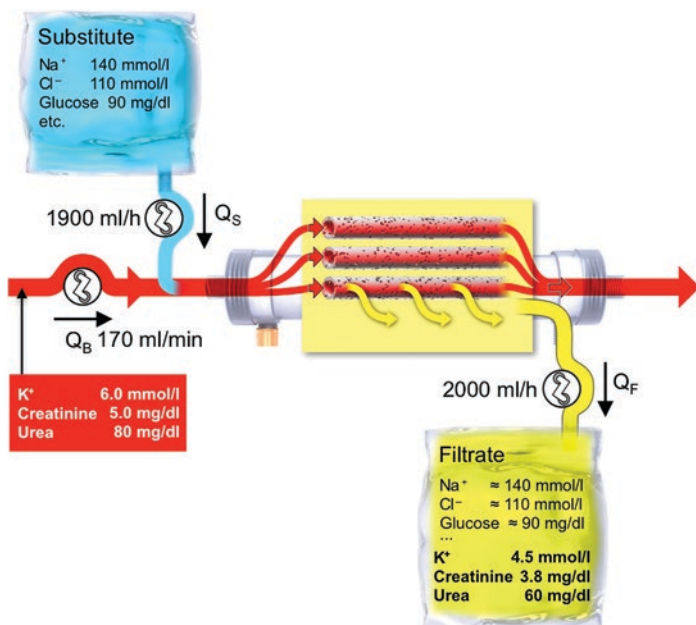
In practice, however, blood flow cannot be increased at will, as even large-lumen venous vascular accesses exhibit physical flow resistance. This flow resistance results in negative and positive pressures in the two limbs of the catheter during blood sampling and return, which in turn are proportional to the blood flow (pressure = flow  $\times$  resistance). In particular, the negative pressure in the leg of the catheter leading away from the patient (“arterial”) leads to haemolysis of the erythrocytes and should therefore be kept as low as possible. Because the blood flow must be limited in certain situations for the reasons mentioned, this also results in upper limits for the filtrate flow  $Q_F$  and thus also for the therapy dose in haemofiltration.

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### 3.5 Predilution: Trade-Off Between Filter Life and Detoxification Performance

High haematocrit values in the filter capillaries and the consequences described can be effectively avoided, if necessary, with a simple modification of veno-venous haemofiltration, which is illustrated in Fig. 3.3. If the substitute is not added to the blood after the filter passage but already added before the haemofilter, a considerable drop in haematocrit occurs due to the “flooding” of the blood. As this diluting effect of the added substitute solution occurs before the filter passage, this variant is referred to as “predilution”—in contrast to the “post-dilution” already described, in which the substitute is added to the blood only after the filter passage.

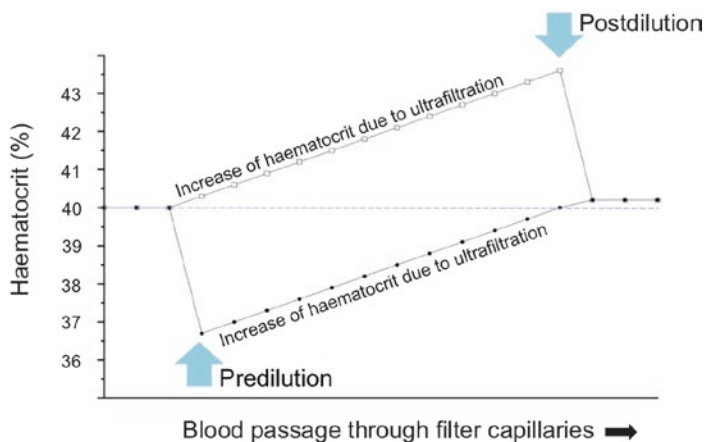
In Fig. 3.4 the course of the haematocrit value during the blood passage through the filter capillaries is qualitatively sketched. At predilution, the haematocrit is almost continuously below the initial value, whereas at post-dilution, it is continuously higher than the initial value and only decreases after the blood has passed through the capillaries due to the addition of the substituent. (As an incidental finding, it should be noted that the haematocrit was slightly higher than the initial value at the end of the filter passage in both procedures. This is the result of net fluid removal ( $Q_S < Q_F$ ), which in most cases should be realized with a renal replacement procedure).



**Fig. 3.3** Continuous veno-venous haemofiltration (CVVH) in predilution mode with typical flow rates for blood, filtrate and substitute. Owing to predilution, the concentrations of the substances to be eliminated are slightly lower in the filtrate than in the blood plasma. (Image creation: MedicalGraphics, Cologne, Germany)

In clinical practice, the predilution variant can actually counteract clotting processes owing to the significantly improved rheology of the blood in the filter capillaries, thereby enabling longer filter service lives. However, the fact that continuous haemofiltration procedures are not generally operated in predilution mode suggests that predilution must also have a disadvantage, namely that the improved rheology in the filter capillaries comes at the price of a slightly reduced efficacy (i.e., a reduced therapeutic dose) of the procedure. Let us consider fluid removal and therapeutic dose for haemofiltration with predilution.

The consideration that fluid is simultaneously withdrawn ( $Q_F$ ) and added ( $Q_S$ ) to the system also applies when using predilution.



**Fig. 3.4** Haematocrit progression in capillaries during haemofiltration in pre- and post-dilution mode. (Own representation)

The net fluid removal is therefore calculated from the difference between the filtrate flow and the substituent flow, just as in the case of a CVVH with post-dilution:

- Haemofiltration with predilution: Fluid removal =  $Q_F - Q_S$

However, because blood and substitution solution already mix *before* the haemofilter, the ultrafiltrate no longer consists exclusively of filtered plasma fluid but also contains more or less large proportions of the previously added substitution solution. The mixing of plasma fluid and substitute upstream of the filter results in lower concentrations of solutes in the filtrate than in the blood plasma (as shown in Fig. 3.3). Thus, a filtration flow  $Q_F$  of 2000 ml/h removes less than 2000 ml/h of plasma fluid and solute dissolved in it.

- Haemofiltration with predilution: Effective therapeutic dose is smaller than  $Q_F$ !

The ratio in which the ultrafiltrate is composed of plasma fluid and substitution solution corresponds to the ratio of the flow rates of plasma fluid and substitute, as these two fluids enter the filter together. If the ratio of blood flow to substitute flow is 5:1 (the

blood flow should always be five times the filtrate flow, and thus also approximately the substitute flow!) and the haematocrit is 40%, the filtrate at predilution approximately consists of 3/4 plasma fluid and 1/4 substitute. The effective therapeutic dose of the procedure is therefore still 75% of the adjusted filtrate flow. This rule of thumb is sufficient for everyday clinical use; for readers interested in mathematics, the exact relationships are shown below:

$$\begin{aligned} &\text{Effective therapeutic dose} \\ &= \text{Filtrate flow} * \frac{\text{Plasma flow}}{\text{Plasma flow} + \text{Substituent flow}} \end{aligned}$$

Plasma flow is calculated from blood flow  $Q_B$  and haematocrit, where division by 100 converts haematocrit from a percentage value to a fraction between 0 and 1, and multiplication by 60 converts plasma flow to ml/h:

$$\text{Plasma flow} = \text{Blood flow} * \left(1 - \frac{Hk}{100}\right) * 60$$

(plasma flow in ml/h, blood flow in ml/min, Hk in %).

Both formulas put together give:

$$\begin{aligned} &\text{Effective therapeutic dose} \\ &= \text{Filtrate flow} * \frac{\text{Blood flow} * \left(1 - \frac{Hk}{100}\right) * 60}{\text{Blood flow} * \left(1 - \frac{Hk}{100}\right) * 60 + \text{Substituent flow}} \end{aligned}$$

(Blood flow in ml/min, filtrate flow and substituent flow in ml/h, Hk in %).

Thus, the higher the substituent flow  $Q_S$  is in relation to the blood flow  $Q_B$  during predilution, the more pronounced is the diluting effect of predilution (i.e., better flow properties of the blood in the capillaries). At the same time, however, a high substituent flow  $Q_S$  also shifts the composition of the filtrate towards a lower proportion of plasma fluid, i.e., the effective therapeutic dose decreases. For clinical application, this means that for hae-

modifiltration in predilution mode, a suitable compromise must always be found between the best possible rheology in the filter capillaries and the lowest possible losses in the therapeutic dose. Blood flow  $Q_B$  and substitute flow  $Q_F$  as determining parameters in predilution procedures must therefore be optimally adapted to the individual medical priorities (filter life vs. effectiveness of therapy) in the treatment of the respective patient.

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### 3.6 Summary of Continuous Veno-Venous Haemofiltration (CVVH)

- In haemofiltration, solutes dissolved in the blood are eliminated by physically squeezing plasma fluid. All molecules that are significantly smaller than the cut-off value of the membrane are eliminated to the same extent, regardless of their size.
- Because the plasma concentration of the vast majority of substances to be eliminated is very low, very large amounts of plasma fluid must be filtered off to achieve quantitatively sufficient elimination of the dissolved substances. To avoid excessive fluid losses, the continuous addition of a substitution solution is required, either before the haemofilter (predilution) or after the haemofilter (post-dilution).
- The “purification” of the blood is achieved by the filtration of plasma fluid and its replacement by an appropriate amount of substitution solution. In post-dilution, the therapeutic dose therefore corresponds to the filtrate flow  $Q_F$ ; in predilution, the therapeutic dose is approximately 75% of the filtrate flow.
- The net fluid removal is the difference between the expired ultrafiltrate and added substituent ( $Q_F - Q_S$ ).
- To avoid excessive haemoconcentration in the filter capillaries, the blood flow  $Q_B$  in post-dilution mode should be at least five times the filtrate flow  $Q_F$  (= therapy dose!). Flow obstacles in the area of the vascular accesses can thus limit the maximum realizable therapy dose!

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- VA/NIH Acute Renal Failure Trial Network, Palevsky P, Zhang J, O'Connor T, Chertow G, Crowley S, Choudhury D (2008) Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med* 359:7–20

# Haemodialysis: Exchange Without Flow

In contrast to haemofiltration, in which solutes are eliminated exclusively via a fluid outflow, the central elimination mechanism of haemodialysis is the diffusion of substances from the blood into a surrounding fluid. For a detailed understanding of this very common modality, the following chapter addresses both the underlying countercurrent principle and the importance of the composition of the dialysate. The “adjusting screws” with which the two treatment parameters of therapy dose and fluid removal can be influenced are discussed, as are the system-related advantages and disadvantages of continuous haemodialysis.

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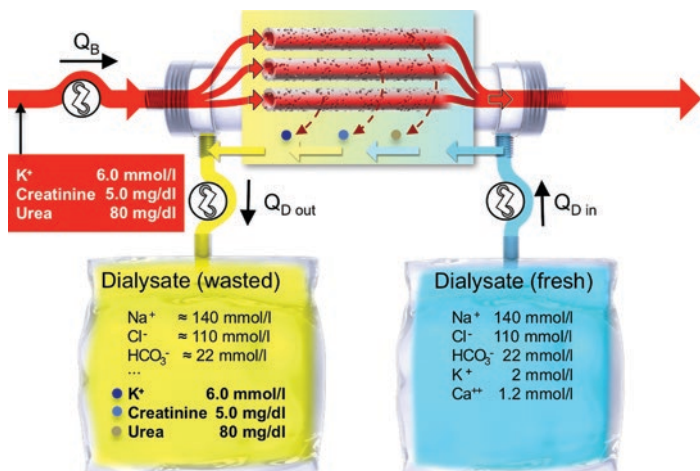
## 4.1 Diffusion as a Basic Principle

After this detailed excursion into the world of continuous renal replacement procedures based on haemofiltration, we return once again to the birth of the first industrially manufactured, sterile and pyrogen-free hollow fibre membranes. The possibility of purifying blood by filtering off plasma fluid was a novel therapeutic approach at the time, which (as described) was refined and further developed by several technical innovations. However, the hollow fibre membranes also made it possible to purify the blood not by filtration but by diffusive mass transfer, thus applying the process



that had already proven itself clinically since the 1950s as “dialysis”: instead of squeezing plasma fluid into the extracapillary compartment of the filter cartridge, thereby eliminating fluid and urinary substances, this compartment is filled with a dialysate fluid. This fluid flows around the blood-permeated microcapillaries so that urinary substances pass from the blood into the dialysate by diffusion without the need for fluid outflow.

Because the dialysate volume inside the filter cartridge is very small (only a few 100 ml), this small amount of dialysate would be saturated with substances diffusing from the blood after a very short time, which would bring any further diffusion to a standstill. For this reason, the extracapillary space of the filter cartridges was provided with a second connection so that fresh dialysate could be continuously fed into the cartridge while the same amount of “used” dialysate (i.e., saturated with substances to be absorbed) flows out the other side. Figure 4.1 shows the schematic structure of continuous veno-venous haemodialysis (CVVHD).



**Fig. 4.1** Schematic structure of continuous veno-venous haemodialysis (CVVHD). Substances requiring elimination have the same concentrations in the outflowing (“used”) dialysate as in the blood plasma. (Image creation: MedicalGraphics, Cologne, Germany)

It can be seen that not only the inflow of dialysate ( $Q_{D \text{ in}}$ ) but also its outflow ( $Q_{D \text{ out}}$ ) is controlled by a separate roller pump. This is imperative because the blood flow also creates a high hydrostatic pressure inside the filter capillaries during dialysis. In the case of “free” outflow, this pressure would cause uncontrolled amounts of plasma fluid to flow out of the capillaries (analogous to the very first, pumpless setup of haemofiltration in Fig. 2.1). However, if both dialysate pumps operate at the same flow rate, the inflow and outflow of dialysate are exactly balanced so that no plasma fluid can flow out of the hollow fibres into the extracapillary space.

This basic technical design of continuous haemodialysis with a hollow fibre membrane filter, a blood pump, an inflowing and an outflowing dialysate pump has remained unchanged over the decades and is still relevant today. Two details are important for the exchange processes between blood and dialysate (i.e., the central process in the purification of blood).

### 4.1.1 Selective Diffusion

In addition to the few urinary substances, blood also contains many dissolved substances that should *not* be removed by dialysis (e.g., sodium, calcium, glucose, bicarbonate). If the most important of these substances are added to the dialysate at physiological concentrations, there are no longer any relevant differences in concentration between blood and dialysate, and there is also no significant diffusion of these substances. The composition of the dialysate therefore determines which substances diffuse from the blood into the dialysate and which do not.

The potassium ion occupies a special position: although potassium is undoubtedly one of the substances that should be eliminated from the blood, most dialysate solutions are not potassium-free but contain potassium in subphysiological concentrations (e.g., 2 mmol/l). This potassium content in the dialysate does reduce the concentration gradient between the blood and dialysate and thus also reduces the extent of potassium diffusion. However, the addition of potassium to the dialysate reliably

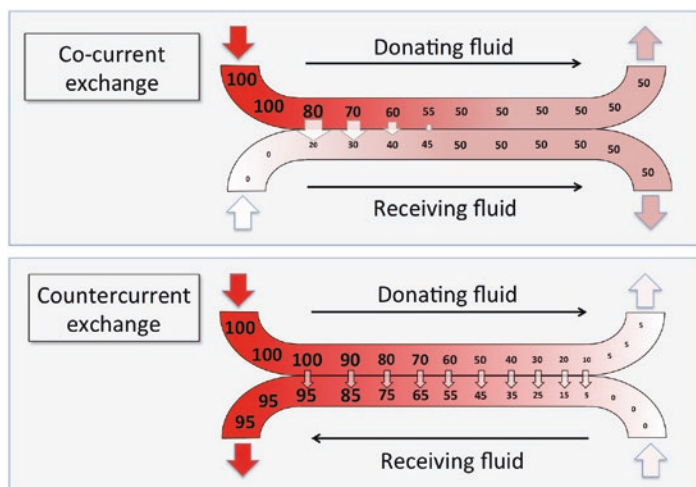
prevents the potassium in the blood from dropping to levels below 2 mmol/l (i.e., life-threatening hypokalaemia), which would in principle be possible with potassium-free dialysate. However, potassium-free dialysate is also available for the treatment of severe hyperkalaemia. When this dialysate is used, a maximum concentration gradient between blood and dialysate results in the greatest possible potassium diffusion. However, for the reasons mentioned above, close monitoring of serum potassium is mandatory in this case.

In addition to the potassium content, dialysate solutions can also differ with regard to the concentration of ionized calcium ( $\text{Ca}^{++}$ ). While the dialysate normally contains  $\text{Ca}^{++}$  in physiological concentrations to prevent  $\text{Ca}^{++}$  from flowing out of the blood,  $\text{Ca}^{++}$ -free dialysate must be used when using citrate anticoagulation. The need for this special composition of dialysate is explained in Sect. 6.2 (How citrate anticoagulation works).

### 4.1.2 Continuous Diffusion in Counterflow Principle

As seen in Fig. 4.1, the delivering fluid (= blood) and the receiving fluid (= dialysate) flow in opposite directions during dialysis: The blood flows from left to right, while the fresh dialysate enters the cartridge via the right port and leaves it as “used” (i.e., contaminant-laden) dialysate via the left port. The countercurrent principle implemented here enables a particularly effective exchange of dissolved substances. Figure 4.2 compares the exchange processes in cocurrent and countercurrent flow.

In the cocurrent process, the dispensing and receiving liquids flow parallel to each other. The concentration of the dissolved substances increases in the receiving liquid and decreases in the discharging liquid. As a result, the concentrations in both liquids quickly equalize and meet approximately in the “middle”; from this point on, no further substance exchange takes place. The absorbing liquid takes over only half of the dissolved substances, and the other half remains in the discharging liquid.



**Fig. 4.2** Exchange processes in the cocurrent and countercurrent processes. In the cocurrent process, approximately half of the substances to be exchanged remain in the dispensing liquid, whereas in the countercurrent process, there is an almost complete exchange of substances between the dispensing and receiving liquid. (Own representation)

In the countercurrent process, the two liquids do not flow parallel to each other but past each other in opposite directions. As with the cocurrent process, the concentration of the substances to be exchanged in the receiving liquid also increases continuously in the countercurrent process. However, as the receiving liquid moves in the direction of the “fresh” discharging liquid, it is constantly opposed by proportions of the discharging liquid that contain the substances to be exchanged in increasing concentrations. As a result, there is a stable concentration gradient between the two liquids during the entire contact so that a continuous exchange of substances takes place. Due to the constantly inflowing discharging liquid, the receiving liquid can saturate to such an extent that the concentration of dissolved substances at the end of the exchange process is almost as high as in the discharging liquid.

In the haemofilter, the blood is the releasing fluid, and the dialysate is the receiving fluid. At the end of the exchange pro-

cess, all substances subject to elimination should have almost the same concentrations in the dialysate as in the plasma fluid. The prerequisite for this complete saturation of the dialysate with substances subject to elimination is that a sufficiently high amount of substances subject to elimination is “delivered” by the blood. In practice, blood flow is therefore usually selected to be 2–3 times as high as dialysate flow. This ensures that the concentrations of substances subject to elimination in the dialysate at the end of the countercurrent passage are almost the same as in the plasma fluid.

- If the blood flow is sufficiently high (i.e., higher than the dialysate flow), the concentrations of substances to be exchanged in the outflowing dialysate are as high as in the plasma fluid.

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## 4.2 Therapeutic Dose of Haemodialysis

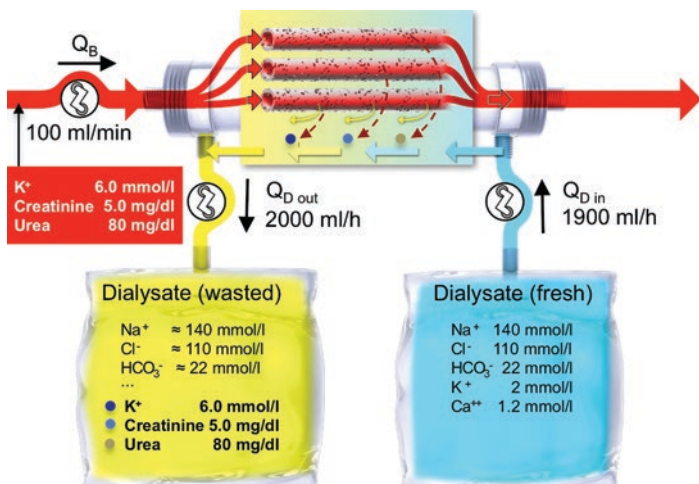
Because the concentrations of the substances to be eliminated (e.g., creatinine, urea, potassium) are the same in the outflowing dialysate as in the plasma water, each litre of outflowing dialysate contains the same amount of contaminants that was previously dissolved in one litre of plasma water. Therefore, each litre of effluent dialysate cleared one litre of plasma fluid of its contaminants. The therapeutic dose of haemodialysis is therefore equal to the dialysate outflow (assuming the blood flow is higher than the dialysate flow). As with haemofiltration, the therapeutic dose for haemodialysis should be in the range of 20–30 ml/kg/h.

- Haemodialysis: therapeutic dose = dialysate outflow rate  $Q_{D \text{ out}}$

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## 4.3 Fluid Removal

The elimination of urinary substances during dialysis requires no fluid outflow, as these substances diffuse from the blood into the dialysate exclusively as a result of a concentration gradient. In most cases, however, the aim of a renal replacement procedure is



**Fig. 4.3** Typical flow rates of continuous veno-venous haemodialysis with fluid removal. (Image creation: MedicalGraphics, Cologne, Germany)

not only to eliminate dissolved substances in the sense of a therapeutic dose but also to generate a net removal of fluid. Fluid removal can also be realized in dialysis procedures with a very simple measure, which is shown in Fig. 4.3.

In the previous considerations, the pumps for the inflow and outflow of the dialysate were operated at the same delivery rates. However, if the dialysate inflow pump is operated at a lower flow rate than the dialysate outflow pump, the outflow pump must fill this gap by withdrawing the missing amount of fluid from the blood in the form of plasma fluid (with all solutes dissolved in it) through the pores of the hollow fibre membrane. The flow rates in Fig. 4.3 illustrate the therapy dose and fluid removal.

The outflowing dialysate pump drains  $2000 \text{ ml/h}$ , but the inflowing dialysate pump only delivers  $1900 \text{ ml/h}$  dialysate. The difference, i.e.,  $100 \text{ ml/h}$ , is filtered out of the blood and is therefore the net fluid removal.

- Continuous haemodialysis: Fluid removal = difference between the delivery rates of the outgoing and incoming dialysate pumps ( $Q_{D \text{ out}} - Q_{D \text{ in}}$ )

This process is indeed ultrafiltration, via which—just as in haemofiltration—all soluble blood components are eliminated. However, a comparison of typical values for dialysate flow (= therapy dose, i.e., approx. 2000 ml/h) and fluid removal (e.g., 100 ml/h) makes it clear that the outflowing fluid consists mainly of dialysate saturated with harmful substances and only to a small extent of filtered plasma fluid. Haemodialysis with removal by filtration therefore does not yet constitute haemodiafiltration. This special form of renal replacement therapy, in which dialysis is combined with a highly effective ultrafiltration component, is discussed in Chap. 5.

Regardless of the ratio of dialysate and ultrafiltrate in the outflow fluid, both the saturated dialysate and the extracted ultrafiltrate contain substances to be eliminated at the same concentrations as the plasma fluid. With every litre of fluid drained off, one litre of blood was freed from its harmful substances. Therefore, also in continuous haemodialysis with fluid removal, the therapy dose corresponds to the running rate of the outflowing dialysate pump.

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#### **4.4 Influence of the Molecule Size on the Diffusion Process**

As already recognized by Thomas Graham, the extent of diffusion of a substance from one compartment to another depends on the concentrations of this substance in both compartments (i.e., the mass of dissolved particles per volume, e.g., in mg/dl). However, the diffusion behaviour of solutes is determined not only by the mass of solute per volume but also by the number of solute particles per volume. Thus, one gram of dissolved small molecules (e.g., potassium) has a much higher diffusion tendency than one gram of large molecules (e.g., myoglobin). Furthermore, the resistance that a molecule has to overcome when moving through a liquid also depends on the size of the molecule; small molecules can move through a liquid much more easily than large molecules.

The general fact that small-molecule substances such as potassium, urea and creatinine diffuse better from the blood into the

dialysate during haemodialysis than large molecules such as creatine kinase (CK) or myoglobin and are therefore eliminated more effectively is derived from these two reasons. Independent of differences in diffusion behaviour, however, the pore size of the membrane also determines up to which molecule size transfer into the dialysate is possible at all. To improve the elimination of large-molecule substances, so-called “high cut-off” filters are available that have a higher cut-off value than conventional dialysis filters. However, their use is associated with the disadvantage of increased diffusion of larger physiological molecules such as albumin or antibodies.

- ▶ Haemodialysis eliminates small-molecule substances better than large-molecule substances. High cut-off filters are available to improve the elimination of large molecular weight substances.

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## 4.5 Summary Continuous Venovenous Haemodialysis (CVVHD)

- In haemodialysis, substances dissolved in the blood are eliminated by diffusing them from the blood into a dialysate solution that flows around the filter capillaries in the countercurrent. This diffusion is not linked to an outflow of plasma fluid from the filter capillaries.
- The inflowing dialysate contains the most important substances that are *not to* be removed from the blood by dialysis (e.g., sodium, glucose, etc.) in physiological concentrations. This effectively prevents the unwanted diffusion of these substances from the blood into the dialysate.
- Small molecules (creatinine, urea, potassium) diffuse more easily into the dialysate and are therefore better eliminated than large molecules such as myoglobin or creatine kinase (CK).
- Because the concentrations of the substances in the dialysate completely match the concentrations in the plasma fluid during countercurrent contact, each litre of outflowing dialysate con-



tains the same amount of solutes as one litre of plasma fluid. The therapeutic dose thus corresponds to the outflow rate of the dialysate.

- If the dialysate inflow pump is operated at a lower flow rate than the dialysate outflow pump, an additional outflow of plasma fluid from the filter capillaries occurs. In addition to the diffusive removal of dissolved particles (which is not bound to any fluid outflow!), fluid removal can also be realized through this.
- In contrast to haemofiltration, blood flow only has to be higher than dialysate flow (= therapy dose); in practice, a ratio of 2:1 to 3:1 is recommended.

## Best of Both Worlds: Haemodiafiltration

After the modalities “haemofiltration” and “haemodialysis” have been dealt with in detail in the preceding chapters, the synthesis of both procedures is discussed in the following chapter: continuous haemodiafiltration. The findings from the previous chapters can easily be integrated into the concept of haemodiafiltration; the derivation of the treatment parameters therapy dose and fluid removal also follows already known considerations.

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### 5.1 From Haemodialysis with Fluid Removal to Haemodiafiltration

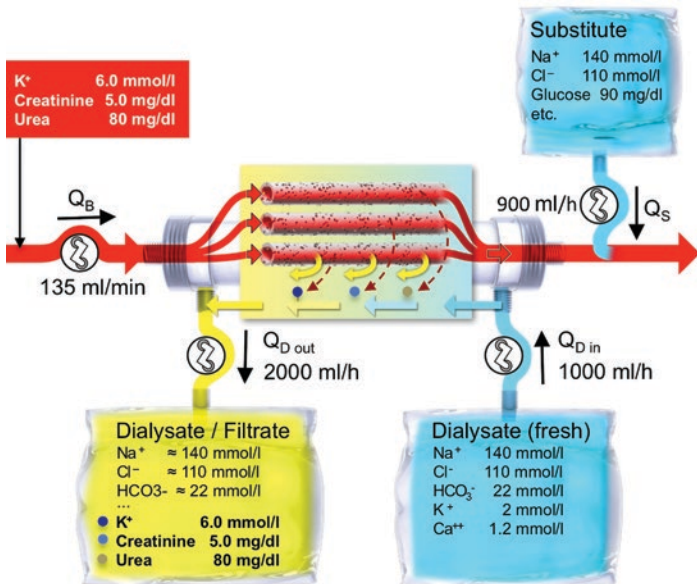
As described in the previous chapter, fluid removal in continuous haemodialysis is realized by the fact that the supplying dialysate pump operates at a slightly lower flow rate than the draining dialysate pump. The draining pump closes this gap by drawing off the missing fluid in the form of plasma water from the microcapillaries.

However, if the supplying dialysate pump runs at a much (!) lower flow rate than the discharging pump, a fluid outflow from the capillaries occurs that is much higher than the actually planned net fluid removal. As already explained in Chap. 2, this is the central concept of haemofiltration. Due to a very high filtrate flow (much higher than necessary for net fluid removal), large amounts

of solutes are eliminated by fluid outflow; for balancing, an almost equal volume of a substitution solution is subsequently added to the blood.

If, in haemodialysis, the two dialysate pumps are operated at a higher run rate difference than would be necessary for net fluid removal, solutes are eliminated both by diffusion into the dialysate and by high-volume filtration. This results in haemodiafiltration, the concept of which is illustrated in Fig. 5.1 using exemplary flow rates.

The draining dialysate pump operates at a flow rate of 2000 ml/h, but the feeding dialysate pump operates only at 1000 ml/h. The difference (1000 ml/h) must be removed from the blood in the form of plasma fluid. Because this would result in a



**Fig. 5.1** Schematic structure of continuous veno-venous haemodiafiltration (CVVHDF) with typical flow rates. Even with the combination of dialysis and filtration, the concentrations of substances requiring elimination in the dialysate/filtrate mixture are the same as in the blood plasma. (Image creation: MedicalGraphics, Cologne, Germany)

fluid loss of 24 l/day, 900 ml/h (i.e., almost the same amount of fluid) is replaced in the form of substitution solution.

- ▶ If more fluid is filtered off than would be required for net fluid removal, haemodialysis becomes haemodiafiltration. Characteristic feature of haemodiafiltration is therefore the addition of a substitution solution.

The two treatment parameters, therapeutic dose and fluid removal, are calculated as follows for haemodiafiltration.

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## 5.2 Therapeutic Dose

The outflowing fluid in Fig. 5.1 consists of two components: 1000 ml/h is dialysate, which was mainly saturated with small-molecule substances by diffusion during the filter passage. The other 1000 ml/h is filtrate, in which all solutes below the cut-off value (i.e., including large molecules) are present at the same concentration as in plasma water. With the exception of large molecules, which are predominantly found in the filtrate, both components contain solutes to be eliminated at the same concentration as plasma water. Regardless of the ratio of dialysate to filtrate in the outflowing fluid, each litre of the outflowing dialysate/filtrate mixture has also always freed one litre of plasma water from its solutes. The therapy dose is therefore the operating rate of the outflowing dialysate pump (which should correctly be called dialysate/filtrate pump), i.e., 2000 ml/h in Fig. 5.1.

Hemodiafiltration : Therapeutic dose = Outflow rate of dialysate /  
filtrate mixture

- ▶ = operating rate of the draining dialysate pump

While the run rate of the outflowing dialysate pump determines the amount of the total therapy dose, the difference between the inflowing and outflowing dialysate flow determines the share of the haemofiltration component in the total therapy dose. The

further the dialysate inflow ( $Q_{D\text{ in}}$ ) lags behind the running rate of the outflowing pump ( $Q_{D\text{ out}}$ ), the more plasma fluid is filtered from the blood, i.e., the filtration component increases.

- Haemodiafiltration: Filtration component of the therapeutic dose =  $Q_{D\text{ out}} - Q_{D\text{ in}}$

This finding can also be illustrated with the help of Fig. 5.1: At the flow rates indicated there, the therapy dose of 2000 ml/h is generated in equal parts by haemodialysis (1000 ml/h) and haemofiltration (1000 ml/h). In contrast, reducing the dialysate inflow to, say, 500 ml/h would result in 1500 ml of plasma fluid being removed from the blood per hour. In this case, the same total therapeutic dose would be divided in a 3:1 ratio (1500 ml/h:500 ml/h) between the two components, haemofiltration and haemodialysis.

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### 5.3 Fluid Removal

In haemodiafiltration, too, the fluid removal ultimately results from the sum of all outflowing fluids minus the sum of all inflowing fluids. In the example in Fig. 5.1, 2000 ml/h leaves the system via the outflowing dialysate pump, while 1900 ml/h is recorded on the inflow side (1000 ml/h inflowing dialysate, 900 ml/h substitute). Balancing all inflows and outflows results in a net fluid removal of 100 ml/h.

- Hemodiafiltration : Fluid removal = Outflows – Inflows  
= Dialysate Outflow – (Dialysate Inflow + Substitute Flow)

Two considerations already made for pure haemofiltration can also be applied to “small haemofiltration”, which is included in every CVVHDF.

Even with CVVHDF, the filtration fraction should not be higher than 20% to avoid haemoconcentration in the filter capillaries. For filtration, the blood flow must therefore be at least five times the filtrate flow (= difference in running rate between the outgoing and incoming dialysate pumps). However, the amount of

blood needed to saturate the dialysate with solutes, i.e., 2–3 times the dialysate inflow, must also be added. With a balanced ratio between dialysate and filtrate flow, the rule of thumb is that the blood flow should be approximately four times the drainage rate of the dialysate/filtrate mixture; in the example in Fig. 5.1, this would be 8000 ml/h or approx. 135 ml/min.

In addition, the substitute can also be added either before or after filtration in a CVVHDF, i.e., a choice between pre- or post-dilution mode must also be made in a CVVHDF. With predilution, the same advantages and disadvantages arise as with pure haemofiltration: improved rheology is offset by reduced effective therapy performance.

A useful application for CVVHDF is in situations where both high molecular weight substances are to be eliminated by filtration and low molecular weight substances by dialysis. An example of this would be pronounced rhabdomyolysis, which releases large amounts of potassium but also creatine kinase (CK) and myoglobin.

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## 5.4 Summary Continuous Venovenous Haemodiafiltration (CVVHDF)

In haemodiafiltration,

- Substances dissolved in the blood are eliminated by two mechanisms: diffusion from the blood into the dialysate and high-volume filtration, which is significantly higher than the desired fluid removal.
- Because the filtration rate is significantly higher than the desired fluid removal, the addition of a substitution solution is necessary. This enables (analogous to pure haemofiltration) a high filtration rate with simultaneously low fluid removal. The substitution can be carried out as pre- or post-dilution.
- The outflowing fluid is a mixture of saturated dialysate and ultrafiltrate. Because both fluids contain substances to be eliminated at the same concentration as plasma water, the therapy

dose corresponds to the outflow rate of the dialysate/filtrate mixture.

- Due to the high-volume filtration component, large molecules such as myoglobin or creatine kinase (CK) are eliminated better than with pure haemodialysis. However, in contrast to haemofiltration, the blood flow only needs to be four times the therapy dose (drainage rate of the dialysate/filtrate mixture).

# If Life Gives You Lemons: Ask for Citrate Anticoagulation

## 6

For the continuous use of renal replacement procedures, effective anticoagulation is required to reliably prevent clotting processes inside the filter capillaries. Citrate anticoagulation has rapidly become the procedure of choice due to its reliable effect and a number of other advantages. Starting with a detailed explanation of all the processes involved in citrate anticoagulation, the following chapter covers all aspects of its practical application. This includes the dosages of citrate and calcium and the interpretation of laboratory parameters but also limitations of citrate anticoagulation in clinical use.

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### 6.1 Anticoagulation in Continuous Renal Replacement Procedures

The implementation of continuous renal replacement therapy requires effective anticoagulation to prevent clotting processes in the filter capillaries. For a long time, systemic anticoagulation with heparin was the gold standard. In contrast to other extracorporeal procedures (e.g., ECMO), renal replacement therapy does not require pTT- or ACT-effective anticoagulation. However, in patients with an increased risk of bleeding, even low-dose heparin



therapy can result in serious complications. In addition, there is an absolute contraindication to the use of heparin in patients with heparin-induced thrombocytopenia (HIT).

Citrate anticoagulation (also known as regional anticoagulation) is an effective, safe and well-controlled alternative to anticoagulation with heparin and has become the standard method of anticoagulation in renal replacement procedures due to its broad applicability.

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## 6.2 Functionality

The concept of citrate anticoagulation is based on the fact that when ionized calcium ( $\text{Ca}^{++}$ , sometimes abbreviated as  $\text{iCa}$ ) falls below 0.5 mmol/l, there is no longer any coagulatory potential in the blood, as  $\text{Ca}^{++}$  is mandatory for both intrinsic and extrinsic coagulation activation.

►  $\text{Ca}^{++} < 0.5$  mmol/l: Blood coagulation no longer present.

Citrate forms very stable calcium-citrate complexes with  $\text{Ca}^{++}$  ions. If approximately 3–4 mmol of citrate per litre of blood is added to the blood before the haemofilter, approximately 3/4 of the  $\text{Ca}^{++}$  ions form a complex with citrate. As a result, the concentration of free  $\text{Ca}^{++}$  drops to values of approximately 0.3 mmol/l so that blood coagulation is de facto suspended. Before the blood is reinfused to the patient (i.e., after the filter passage), the appropriate amount of free  $\text{Ca}^{++}$  is added to the blood to raise the  $\text{Ca}^{++}$  concentration back to physiological levels (approximately 1.2 mmol/l) and completely abolish hypocalcaemia-induced anticoagulation. The advantages of citrate anticoagulation are as follows:

- Anticoagulation begins with the addition of citrate and ends with the substitution of  $\text{Ca}^{++}$ . It is thus limited exclusively to the extracorporeal blood stock in the haemofilter (hence the term “regional anticoagulation”); the patient has the full coagulatory potential of his blood available in vivo. This is highly

advantageous in many clinical situations (intracranial or retro-peritoneal bleeding, extensive trauma, etc.).

- Citrate is a physiological, ubiquitously occurring molecule in metabolism. Allergies or intolerances (such as HIT) therefore do not occur.
- The formation of calcium-citrate complexes occurs independently of the organ and according to stoichiometric laws. Citrate anticoagulation is therefore easily controllable; comparable effects, such as occasionally observed heparin resistance, do not exist with citrate anticoagulation.

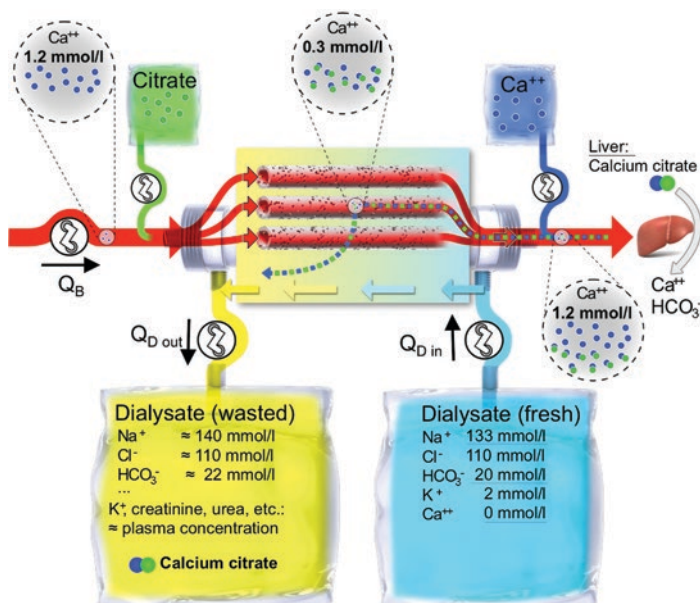
Citrate anticoagulation can be used in all renal replacement procedures (CVVH, CVVHD, CVVHDF, pre- or post-dilution). The basic structure of a renal replacement procedure with citrate anticoagulation is shown in Fig. 6.1 using the example of continuous haemodialysis (CVVHD).

Familiar structures can be recognized in the core: On the left of the image is the blood pump that generates the blood flow  $Q_B$ , below it is the dialysate pump with a dialysate outflow  $Q_{D\text{ out}}$ , and opposite it is the dialysate pump with the delivery rate  $Q_{D\text{ in}}$ .

Between the blood pump and the haemofilter, a fourth pump adds 3–4 mmol of citrate per litre of blood, binding approximately 3/4 of the ionized calcium in the form of calcium-citrate complexes. This causes the  $\text{Ca}^{++}$  concentration in the capillaries to drop to values below 0.3 mmol/l so that blood coagulation in the filter is effectively inhibited.

► Citrate dose: approximately 3–4 mmol/l of blood.

Calcium-citrate complexes are very small molecules. Because the dialysate does not contain any calcium-citrate complexes, approximately half of the complexes formed diffuse from the blood into the dialysate during the filter passage and are eliminated from the system. The other half of the calcium-citrate complexes enter the liver with the bloodstream and are degraded to  $\text{Ca}^{++}$  and bicarbonate.



**Fig. 6.1** Functional principle of citrate anticoagulation using the example of continuous veno-venous haemodialysis (CVVHD). The addition of citrate causes the concentration of  $\text{Ca}^{++}$  in the capillaries to fall to values of approximately 0.3 mmol/l, and coagulation is thus abolished. After the filter passage, the  $\text{Ca}^{++}$  concentration is raised again to normal values by the addition of  $\text{Ca}^{++}$ . The calcium-citrate complexes either diffuse into the dialysate or are degraded by the liver to  $\text{Ca}^{++}$  and bicarbonate. (Image creation: Medical-Graphics, Cologne, Germany)

As seen on close inspection in Fig. 6.1, the fresh dialysate differs from the “conventional” dialysate in three respects when citrate anticoagulation is used:

- The dialysate does not contain  $\text{Ca}^{++}$ . This difference (already briefly mentioned in Sect. 4.1.1) is easy to understand, A physiological  $\text{Ca}^{++}$  concentration of approximately 1.2 mmol/l in the dialysate would result in a permanent influx of  $\text{Ca}^{++}$  into the capillaries due to the much lower  $\text{Ca}^{++}$  concentration inside the capillaries (approximately 0.3 mmol/l). This would attenuate

or completely abolish citrate-mediated hypocalcaemia and thus counteract anticoagulation. To prevent this, the dialysate for citrate anticoagulation does not contain  $\text{Ca}^{++}$ .

- The concentration of bicarbonate in the dialysate is slightly subphysiological at 20 mmol/l, so that small amounts of bicarbonate are shifted from the blood into the dialysate and eliminated. This compensates for the continuous accumulation of bicarbonate that occurs during hepatic degradation of calcium-citrate complexes. Normally, citrate anticoagulation is therefore pH neutral.
- The sodium concentration in the dialysate is also slightly subphysiological at approximately 133 mmol/l, so that even a small amount of sodium is continuously diffused out of the blood into the dialysate and eliminated. This compensates for the sodium influx that occurs with citrate anticoagulation because citrate molecules are present as sodium salts, and thus, a certain amount of sodium is always added to the blood.

After the blood has passed through the haemofilter, the amount of ionized calcium that has previously diffused from the blood into the dialysate in the form of calcium-citrate complexes and was thus removed from the patient is added to the blood before it is returned to the patient venously. The returning blood contains  $\text{Ca}^{++}$  at a physiological concentration of approximately 1.2 mmol/l, plasmatic coagulation is thus no longer impaired in the patient. The dosage of calcium substitution is discussed in more detail in Sect. 6.5.

When using regional citrate anticoagulation, there are three key aspects that need to be monitored regularly (i.e., approximately every 8 h).

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### 6.3 Effectiveness of Anticoagulation

For complete abolition of plasmatic coagulation, the  $\text{Ca}^{++}$  concentration in the filter capillaries must be kept within a range of 0.25–0.35 mmol/l. Blood sampling to check the current  $\text{Ca}^{++}$  con-

centration is understandably not possible directly from the filter capillaries due to their microstructure. However, as long as no  $\text{Ca}^{++}$  has been added to the blood after the filter passage, blood can be taken after the filter whose  $\text{Ca}^{++}$  concentration corresponds to that in the filter. For this reason, dialysis systems have a blood sampling port immediately downstream of the filter from which blood is taken for the measurement of the so-called “postfilter”  $\text{Ca}^{++}$ .

- “Postfilter calcium” =  $\text{Ca}^{++}$  concentration in the filter, target range 0.25–0.35 mmol/l

If the postfilter  $\text{Ca}^{++}$  exceeds the limit of 0.35 mmol/l, sufficient anticoagulation may no longer be given, in which case the citrate dosage (given in mmol citrate per litre of blood) must be increased. This is usually achieved by increasing the citrate flow but reducing the blood flow would have the same effect with regard to the concentration of free  $\text{Ca}^{++}$ . Conversely, the amount of citrate added per litre of blood must be reduced if the postfilter  $\text{Ca}^{++}$  falls to values below 0.25 mmol/l.

- Postfilter  $\text{Ca}^{++} > 0.35$  mmol/l: increase citrate dose (mmol citrate per litre blood).
- Postfilter  $\text{Ca}^{++} < 0.25$  mmol/l: reduce citrate dose (mmol citrate per litre blood).

---

## 6.4 Blood pH

As previously described, approximately half of the calcium-citrate complexes formed enter the liver with the bloodstream, where they are degraded to  $\text{Ca}^{++}$  and bicarbonate. Thus, citrate anticoagulation continuously produces small amounts of bicarbonate. This is counteracted by the fact that the bicarbonate concentration in dialysate solutions for citrate anticoagulation is somewhat lower than in conventional dialysate solutions, which means that a small amount of bicarbonate continuously diffuses from the blood into the dialysate. However, as these two effects do

not always have to balance each other out completely, the patient's blood pH is the second parameter to be monitored regularly.

### 6.4.1 Metabolic Acidosis

An acidotic metabolic state can be counteracted by allowing more calcium-citrate complexes to reach the liver, where they are metabolized to bicarbonate. This can be achieved in two ways: either blood or citrate flow are increased (the citrate dose in mmol per litre of blood remains the same!), so that more calcium-citrate complexes are formed. The other option is to reduce the dialysate flow while keeping the blood and citrate flow constant so that a smaller proportion of the calcium-citrate complexes formed diffuse into the dialysate and are washed out. However, it should be noted that this approach reduces the therapy dose (= dialysate flow!).

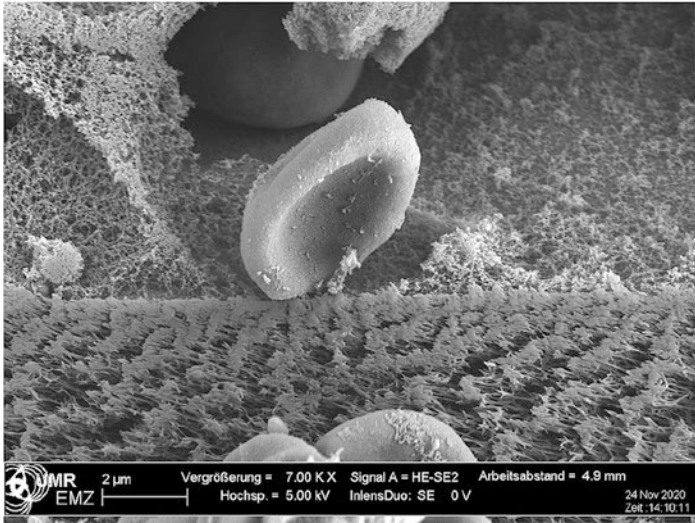
- Metabolic acidosis: increase blood and citrate flow or decrease dialysate flow.

### 6.4.2 Metabolic Alkalosis

Analogous to the abovementioned considerations, an alkalotic metabolic situation occurs when too many calcium-citrate complexes reach the liver with the blood and are degraded to bicarbonate there. This can be counteracted by reducing the blood and citrate flow (= formation of fewer calcium-citrate complexes) or by increasing the dialysate flow (increasing the elimination of calcium-citrate complexes via the dialysate).

- Metabolic alkalosis: Reduce blood and citrate flow or increase dialysate flow.

In addition to the effects of blood, dialysate and citrate flow on blood pH mentioned above, there is another mechanism by which (usually severe) metabolic alkalosis can develop: coagulation pro-



**Fig. 6.2** Coagulation processes inside a microcapillary. In longitudinal section, fibrin nets lining the inner wall of the capillary can be seen at a magnification of 7000 $\times$ . This impairs both the diffusion of solutes from the blood into the dialysate and the filtration of plasma fluid. (Source: Electron Microscopy Center of the University Hospital Rostock, Germany)

cesses inside the capillary can cause a dense network of fibrin and other molecules to form on the inside of the hollow fibre membranes (see Fig. 6.2). As a result, the small pores of the membrane are obstructed from the inside so that hardly any calcium-citrate complexes can diffuse from the blood into the dialysate. The amount of calcium-citrate complexes reaching the liver thus multiplies rapidly, resulting in the formation of very large amounts of bicarbonate. Increases in the base excess (BE) on the order of 10 mmol/l per hour and an extremely alkaline blood pH are the result; the measures described for metabolic alkalosis (reduction of blood and citrate flow, increase in dialysate flow) have almost no effect in this case.

In the case of a strong increase in base excess that cannot be explained in any other way, the possibility of pore occlusion of the

hollow fibre membranes should therefore be considered. In addition to severe metabolic alkalosis, this process also results in a severe decrease in the diffusion of all other substances requiring elimination from the blood into the dialysate. Despite a formally adequate dialysate flow, the effective cleansing performance of the therapy procedure thus decreases considerably, which is why the complete change of the entire system is usually unavoidable in such a situation.

In addition to the characteristic effects on the acid-base balance, increasing obstruction of the pores in the hollow fibre membranes can also be recognized by an increase in the so-called transmembrane pressure. The pressures in a renal replacement system and their interpretation are discussed in Chap. 7.

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## 6.5 Systemic Calcium

Citrate anticoagulation is accompanied by a considerable turnover of calcium: as already described, approximately 3/4 of the free  $\text{Ca}^{++}$  is bound in the form of calcium-citrate complexes, of which approximately half is removed via the dialysate. To fully compensate for this calcium loss via the dialysate, the same amount of calcium must be added back to the blood after filter passage in the form of  $\text{Ca}^{++}$  substitution. Due to the high turnover, elimination and substitution must match exactly; even small percentage deviations between elimination and substitution result in a noticeable increase or decrease in the patient's systemic calcium.

Calcium citrate complexes behave like all other solutes in the blood during dialysis: the extent of their elimination is determined exclusively by the level of dialysate flow. The amount of calcium substitution thus depends directly on the dialysate flow, and the substitution dose is therefore expressed in mmol  $\text{Ca}^{++}$  per litre of dialysate. Under typical conditions (75% of the  $\text{Ca}^{++}$  forms complexes, 50% of the calcium-citrate complexes are dialyzed), the  $\text{Ca}^{++}$  substitution is in the range of 1.5–2.0 mmol  $\text{Ca}^{++}$  per litre of dialysate flow.



- $\text{Ca}^{++}$  substitution: Approximately 1.5–2.0 mmol/l of outflowing dialysate.

For the precise adjustment of  $\text{Ca}^{++}$  substitution, regular measurement of systemic  $\text{Ca}^{++}$  (i.e., free  $\text{Ca}^{++}$  in the patient's blood) is required; this must lie within the range of physiological normal values, i.e., between 1.15 and 1.35 mmol/l. This value is sometimes referred to as “prefilter calcium” when using citrate anticoagulation. The term “prefilter calcium” is confusing because the blood for the determination of the systemic  $\text{Ca}^{++}$  can of course be taken from any (central) venous or arterial catheter and does not have to be taken directly before the filter. The term prefilter  $\text{Ca}^{++}$  for the  $\text{Ca}^{++}$  concentration in the patient's blood should therefore rather be understood as a terminological analogy to postfilter  $\text{Ca}^{++}$ , which reflects the  $\text{Ca}^{++}$  concentration in the filter.

If the patient's systemic  $\text{Ca}^{++}$  is below the normal range, the  $\text{Ca}^{++}$  substitution (expressed in mmol  $\text{Ca}^{++}$  per litre of dialysate) must be increased—and vice versa.

- Systemic  $\text{Ca}^{++} < 1.15$  mmol/l: Increase  $\text{Ca}^{++}$  substitution.
- Systemic  $\text{Ca}^{++} > 1.35$  mmol/l: reduce  $\text{Ca}^{++}$  substitution.

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## 6.6 Addition of Citrate: A Kind of Predilution

The amount of citrate that has to be added to the blood before the filter passage can be easily calculated from the blood flow and the citrate dose. For example, with a blood flow of 100 ml/min (= 6 l/h) and a citrate dose of 4 mmol/l of blood, 24 mmol of citrate per hour must be added to the blood.

For a given amount of citrate per unit time, the lower the citrate concentration in the solution used, the higher the running rate of the citrate solution (and vice versa). This relationship is of practical importance because citrate solutions of widely varying concentrations are available: At the time this book went to press, citrate solutions with concentrations ranging from 10 to 136 mmol/l were being prepared.

To add 24 mmol of citrate per hour to the blood, the running rate of a highly concentrated citrate solution (136 mmol/l) would have to be approximately 176 ml/h, while a running rate of 2400 ml/h would be required for the same dose when using a low-concentration citrate solution (10 mmol/l). However, to make the citrate anticoagulation procedure balance neutral, an equal amount of fluid must always be removed from the blood during filter passage by ultrafiltration.

The constellation of fluid addition before filter and fluid removal in the filter is reminiscent of haemofiltration in the predilution mode (Sect. 3.5 and Fig. 3.3). Here, too, a large amount of fluid is removed from the blood by ultrafiltration and an almost equal amount of fluid is added in the form of a substitution solution before the filter passage.

However, in haemofiltration (CVVH) in predilution mode and citrate anticoagulation, the “causalities” of fluid withdrawal and fluid addition are exactly reversed: in haemofiltration, high-dose filtration is the primary process for the elimination of solutes, while the addition of the substitution solution merely compensates for the high fluid loss associated with filtration. In contrast, for citrate anticoagulation, the addition of the citrate solution is the therapeutic process for anticoagulation, whereas the subsequent filtration compensates for the resulting fluid intake. Although filtration of plasma fluid during citrate anticoagulation serves only to compensate for a previous fluid intake, it nevertheless causes elimination of solutes from the blood just as effectively as during haemofiltration.

As has already been estimated, the citrate supply and compensatory filtration when using a highly concentrated citrate solution (136 mmol/l) at usual blood flows are in the range of approximately 100–200 ml/h. However, filtration in this range is far below the required therapeutic dose of 20–30 ml/kg/h, so additional “purification mechanisms” are required. In most cases, this is realized by an appropriate dialysate flow, which takes over the necessary purification capacity (see Fig. 6.1). Despite minor filtration, this setup would be pure continuous haemodialysis (CVVHD).

Alternatively, however, the desired therapy dose can also be realized via a significantly higher filtration rate and the addition of an additional, citrate-free substitution solution after the filter. In this case, the renal replacement procedure would be pure continuous haemofiltration (CVVH), and substitution would take place both as predilution (citrate solution) and post-dilution (“classic” substitution solution).

The situation is completely different when using a low-concentration citrate solution. With a citrate concentration of 10 mmol/l and a citrate requirement of 24 mmol/h, the citrate solution must be supplied to the blood at a flow rate of 2400 ml/h. To compensate for this inflow, a volume of 2400 ml/h must be withdrawn from the blood via the mechanism of filtration. In a normal-weight patient, this corresponds to a filtration rate of approximately 35 ml/kg/h so that effective blood purification is achieved by compensating for the addition of citrate alone. The citrate solution therefore has two functions: on the one hand, it ensures adequate anticoagulation, and on the other hand, it assumes the function of the substitution solution, through which a filtration of more than 2 l/h and a significantly lower withdrawal become compatible with each other. When using low-dose citrate solutions, the resulting mode is therefore always haemofiltration or haemodiafiltration due to the high filtration turnover; pure haemodialysis is not possible with these citrate solutions.

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## **6.7 Citrate Anticoagulation in Liver Failure**

The liver plays a central role in citrate anticoagulation: it degrades the calcium-citrate complexes that return to the patient with the blood to  $\text{Ca}^{++}$  and bicarbonate. Incomplete degradation leads not only to an increase in calcium-citrate complexes in the blood but also to a reduced formation of  $\text{Ca}^{++}$  and bicarbonate. The first signs of so-called “citrate accumulation” are therefore an increasing  $\text{Ca}^{++}$  substitution requirement and a lowered serum bicarbonate with metabolic acidosis. Although the concentration of calcium-citrate complexes in the blood cannot be determined directly by laboratory chemistry, complex-bound calcium is detected by mea-

asuring the total calcium (total calcium = sum of nonionized calcium and  $\text{Ca}^{++}$ , normal value approx. 2.2–2.5 mmol/l). An increase in total calcium is thus the third sign of incomplete degradation of the calcium-citrate complexes formed.

► Typical triad in the presence of “citrate accumulation”:

- Increasing  $\text{Ca}^{++}$  substitution requirements.
- Drop in serum bicarbonate, metabolic acidosis.
- Increase in total calcium.

Another indicator of citrate accumulation as a result of incomplete degradation of calcium-citrate complexes is the ratio of total calcium to ionized calcium ( $\text{Ca}^{++}$ ).

Under physiological conditions, approximately half of the calcium stock is present in ionized form, so the ratio of total calcium to  $\text{Ca}^{++}$  is approximately 2:1. If the conversion of calcium-citrate complexes to  $\text{Ca}^{++}$  no longer takes place completely, this ratio shifts in the direction of total calcium; the quotient of total calcium and  $\text{Ca}^{++}$  increases. If the ratio of total calcium to  $\text{Ca}^{++}$  increases to values  $\geq 2.5$ , the continuation of citrate anticoagulation in patients with impaired liver function must be critically questioned.

► Ratio total calcium/ $\text{Ca}^{++} \geq 2.5$ : citrate accumulation!

However, especially in patients with hepatic insufficiency, citrate anticoagulation is an interesting alternative to systemic anticoagulation, as impairment of liver function is often associated with coagulation disorders, and systemic anticoagulation should therefore be avoided if possible. The applicability of citrate anticoagulation in patients with impaired liver function has been investigated in a large number of clinical trials. In the vast majority of studies, citrate accumulation was shown to occur rarely, even in the setting of advanced hepatic insufficiency. A meta-analysis summarized the results of ten clinical trials to the effect that liver dysfunction per se is not an exclusion criterion for the use of citrate anticoagulation (Zhang et al. 2019). However, for the early detection of citrate accumulation, continuous attention must be paid to the development of meta-

bolic acidosis and increasing  $\text{Ca}^{++}$  substitution requirements in these cases. The ratio of total calcium to ionized  $\text{Ca}^{++}$  should also be determined regularly.

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## 6.8 Summary of Citrate Anticoagulation

- By adding citrate, the  $\text{Ca}^{++}$  concentration in the blood is greatly reduced, and the blood coagulation in the filter is completely eliminated.
- The reference parameter for the extent of anticoagulation in the filter is the postfilter  $\text{Ca}^{++}$ . The target range is between 0.25 and 0.35 mmol/l and is controlled by the amount of citrate added to the blood before the filter
- After filter passage, anticoagulation is completely antagonized by adding to the blood the amount of  $\text{Ca}^{++}$  previously eliminated in the form of calcium-citrate complexes. The target parameter for the dosage of the substituted calcium is the systemic  $\text{Ca}^{++}$  (or “prefilter”  $\text{Ca}^{++}$ )
- An increase in blood flow leads to an alkalotic metabolic state, whereas an increase in dialysate flow shifts the blood pH towards acidotic values (and vice versa in each case!).
- In the case of impaired liver function, attention must be paid to metabolic acidosis and an increasing need for  $\text{Ca}^{++}$  substitution. The ratio of total calcium to ionized  $\text{Ca}^{++}$  must be monitored closely.

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## Reference

Zhang W, Bai M, Yu Y, Li L, Zhao L, Sun S, Chen X (2019) Safety and efficacy of regional citrate anticoagulation for continuous renal replacement therapy in liver failure patients: a systematic review and metaanalysis. Crit Care 23:22. <https://doi.org/10.1186/s13054-019-2317-9>

# It Is Not Always Just About Flow: Pressure Parameters of Renal Replacement Therapy

During the application of continuous renal replacement therapies, pressures are measured at various points. These pressures (including their time course) provide important information about the condition of the entire system, which is why an understanding of these pressures is helpful in practical applications. The penultimate chapter of this book is therefore devoted to the interpretation of pressure values in a renal replacement system.

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## 7.1 Pressure, Flow and Resistance

While the focus in previous chapters has been on the *flows* of blood, dialysate, filtrate and other substances in a renal replacement system, this penultimate chapter will deal with pressures in a renal replacement system. Pressure and flow are inextricably linked, and the interaction of these two variables can be viewed from two sides. On the one hand, it is always a pressure that acts as a driving force to make a fluid flow against a resistance, i.e., pressure produces flow against resistance. However, if a fluid flows against a resistance (e.g., the water in a garden hose against a narrow nozzle at its end), there is a congestion and thus again a pressure, i.e., flow against resistance generates pressure. Regardless of whether pressure is considered the *cause of* flow or the *result of* flow, pressure always results from the product of flow and resistance.

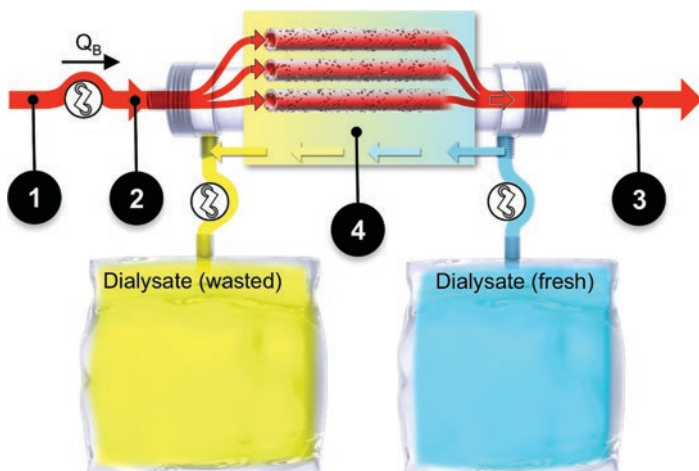
► Pressure = flow \* resistance

All flow rates in renal replacement devices are generated with high accuracy by roller pumps and remain constant as long as the therapy parameters are not changed. Unusually high pressures or sudden changes in a measured pressure are therefore in most cases due to an increase in the resistance against which the fluid must flow.

In renal replacement devices, blood, dialysate and other fluids flow through tubes that can be quite long (the blood lines to and from the patient, for example, are approximately 2 m long). However, the resistance of these tubes is negligible because they have a comparatively large diameter. As a reminder, the flow resistance of a tube decreases with the 4th power of the diameter. If the diameter of a hose is doubled, its resistance is reduced to  $1/(2^4) = 1/16$  of the initial value.

Significant resistances for the blood flow therefore only exist at two points. First, the microcapillaries of the haemofilter represent a relevant flow resistance for the blood due to their very small diameter, which may well increase further during therapy due to clotting processes inside the hollow fibre membrane. The other resistance lies in venous vessel access for the withdrawal and return of the blood: although the catheters used have a comparatively large lumen, the inlet and outlet openings at the catheter tip are effective “bottlenecks” from a hydrodynamic point of view. In addition, the resistance of vascular access can also increase significantly, e.g., due to kinking of the catheter or contact with the inner wall of the vessel.

In modern renal replacement devices, pressures are measured at four different points, which are considered below (see Fig. 7.1). The figure shows continuous haemodialysis, but all considerations regarding pressures also apply to haemo(dia)filtration, as the physical relationships of pressure, flow and resistance are the same for all renal replacement procedures.



**Fig. 7.1** Pressure measurement points in a renal replacement device using the example of continuous veno-venous haemodialysis (CVVHD). (Image creation: MedicalGraphics, Cologne, Germany)

### 7.1.1 Arterial Pressure

The first pressure measurement is taken upstream (i.e., on the patient side) of the blood pump, which draws blood from the patient at the selected flow rate  $Q_B$  (pressure 1 in Fig. 7.1).

As with any blood collection from a venous catheter, negative pressure is required for this procedure. The pressure *before* the blood pump is therefore always negative; the level of this negative pressure depends on the blood flow rate and the resistance that venous vascular access offers to blood flow. An acute increase in this negative pressure (i.e., towards more negative values) at constant blood flow is an indication of flow obstructions at the venous catheter, e.g., due to thrombosis, kinking, etc.

Erythrocytes are exposed to high pressures during each passage through the arterial system of the blood circulation; therefore, they are by nature very stable against overpressure. Negative pressure, on the other hand, leads relatively quickly to



the bursting of erythrocytes, i.e., haemolysis. Aspiration-induced negative pressure is therefore of particular importance; if values are continuously high, it is advisable to check haemolysis parameters (e.g., free haemoglobin) regularly. To reduce the negative pressure, either the blood flow rate can be reduced, or attempts can be made to reduce the flow resistance of venous vascular access.

The terminus “arterial” pressure is misleading, since at first glance it does not necessarily seem understandable that an “arterial” pressure is measured in a veno-venous renal replacement procedure and that this pressure also has negative values. However, as mentioned in Chap. 2, for historical reasons, blood inflow is also referred to as the “arterial” connection in veno-venous procedures. Since the blood on the “arterial” side can only be withdrawn by negative pressure, this “arterial” pressure always has a negative sign.

### 7.1.2 Prefilter Pressure

While the blood pump generates a negative pressure on the patient side to withdraw blood from the patient, an overpressure is generated on the “output side” of the pump, since the blood must flow against the capillary resistance of the haemofilter and the flow resistance of the returning venous vessel access until it is returned to the patient. This pressure is measured before the actual haemofilter (pressure 2 in Fig. 7.1), which is why it is called prefilter pressure.

The prefilter pressure is caused by the fact that blood has to flow serially against two resistances. There are therefore two possible causes for an increase in prefilter pressure with constant blood flow: either the flow resistance of the haemofilter may have increased due to increasing obstruction of the microcapillaries, or there may be a stenosis in the returning vessel access, e.g., due to kinking or thrombosis. In both cases, an increase in pressure upstream of the haemofilter results; to differentiate between the two possible causes, further pressure is required, which is described in the next section.

### 7.1.3 Venous Pressure

The third pressure measurement is taken distal to the haemofilter before the blood flows back into the patient via venous vascular access (pressure 3 in Fig. 7.1). Because the blood flow at this point only has to overcome the flow resistance of venous vascular access, the pressure at this point depends only on the level of blood flow and the resistance of the returning vascular access. With unchanged blood flow, an increase in this pressure therefore immediately indicates increased resistance in the returning vascular access.

Venous pressure also allows a differentiated interpretation of an increased prefilter pressure. If the prefilter pressure is increased but the venous pressure is normal, the stenosis must be located “upstream” of the venous pressure measurement point, i.e., in the haemofilter. However, if both the venous pressure and the prefilter pressure are elevated, the stenosis is in the vascular access. In this case, the increased return pressure will only propagate upstream, causing an increase in prefilter pressure.

► Increased prefilter pressure:

- Venous pressure normal: Increased resistance in the haemofilter
- Venous pressure increased: Increased resistance in the returning vessel access

### 7.1.4 Ultrafiltrate Pressure and Trans-membrane Pressure

The three pressures considered thus far result from a *bloodstream* flowing against two defined resistances. However, there is another fluid flow that also has to overcome a resistance. When fluid is removed from the blood via the mechanism of filtration, this plasma fluid flows through micropores from inside the capillaries into the extracapillary space of the filter cartridge. Owing to their very small diameter, these micropores also exert a relevant resistance to the fluid outflow, so that a defined pressure is required to filter fluids out of the hollow fibre membranes.

On closer inspection, however, the driving force for the fluid outflow is not only the pressure inside the filter capillaries but the pressure *difference* between the inside and the outside of the capillaries. If the fluid outside the capillaries was under the same pressure as the blood inside the capillaries, there would be no fluid outflow at all. The pressure that the blood is under inside the capillaries is known. This pressure corresponds to the prefilter pressure at which the blood pump pumps the blood into the hollow fibre system. To determine the pressure difference between the two sides of the capillary membrane, the fourth pressure to be measured is the pressure outside the hollow fibre membranes, i.e., in the dialysate or ultrafiltrate compartment of the filter cartridge (pressure 4 in Fig. 7.1). In pure haemofiltration, this compartment contains the squeezed ultrafiltrate, whereas in haemodialysis, it contains the dialysate fluid. However, since the fluid is also removed by filtration during haemodialysis, the term ultrafiltrate pressure has become common for the pressure inside the filter cartridge.

- Ultrafiltrate pressure = pressure in the extracapillary compartment of the filter cartridge.

The pressure difference between the inside and outside of the hollow fibre membrane is therefore the difference between the prefilter pressure and the ultrafiltrate pressure. This pressure difference is called transmembrane pressure (abbreviation: TMP).

- Transmembrane pressure (TMP) = prefilter pressure – ultrafiltrate pressure

According to the relationship  $\text{pressure} = \text{flow} * \text{resistance}$ , the TMP depends on the level of filtrate flow and the resistance of the capillary pores (not to be confused with the flow resistance that the capillary lumen opposes the blood flow!). However, unlike blood flow, which varies only within relatively narrow limits, filtrate flow can vary widely with different modalities of renal replacement therapy. In haemodialysis, only the amount of fluid that is actually to be removed from the patient is filtered off, i.e., rarely more than 300 ml/h. In haemofiltration, on the other hand,

the entire blood is cleansed by the mechanism of filtration (with subsequent addition of substitute), so the filtration rate is 20–30 ml/kg/h, i.e., approximately 1500–3000 ml/h and thus a good 10 times that of haemodialysis. In haemodiafiltration or haemofiltration, the higher filtrate flows therefore result in much higher transmembrane pressures than in haemodialysis.

Regardless of its absolute level, an increase in TMP with unchanged filtrate flux indicates that resistance to fluid outflow from the capillaries is increasing, i.e., obstructing the micropores of the hollow fibre membrane by clotting, protein deposition and similar processes. Figure 6.2 shows, for example, a dense fibrin network on the inside of the cut capillary, which largely obstructs the pores of the membrane. In dialysis, this means that there is hardly any diffusion of substances to be eliminated into the dialysate, i.e., the effectiveness of the therapy decreases despite adequate dialysate flow. In the case of citrate anticoagulation, the reduced diffusion of calcium-citrate complexes into the dialysate also leads to a pronounced metabolic alkalosis, as already described in Sect. 6.4.2. Any increases in TMP that are not due to an increase in the filtration rate should therefore be evaluated critically, as the increasing obstruction of the capillary pores has far-reaching consequences for the effectiveness of the procedure and, in the case of citrate anticoagulation, also for the acid-base balance.

This final chapter summarizes the key decision-making processes that need to be made before starting and during continuous renal replacement therapy. It is based on the key findings from the preceding seven chapters of this book (the reading of which is, however, recommended despite this summary). As a kind of “checklist”, the final pages of this book help to ensure that nothing is left to chance when planning and implementing a continuous renal replacement procedure.

---

## 8.1 Who Needs Which Renal Replacement Procedure and When?

*Clinical* recommendations on the use of renal replacement procedures (like all medical knowledge) are subject to constant change. Whether a renal replacement procedure should be started early or delayed as long as possible, how long it should be used, what alternative options exist for a “relative” indication for a renal replacement procedure, and many other clinical decisions can only be made on the basis of current, evidence-based findings.

However, if the indication for a renal replacement procedure is given in a specific situation, the details of the treatment should not be prescribed according to the widespread “as usual” scheme but should be optimally adapted to the patient’s individual situation. The decision paths for the selection, prescription and implementation of a continuous renal replacement procedure are briefly summarized again in the conclusion.

---

## **8.2 Which Modality (Haemodialysis, Haemofiltration or Haemodiafiltration)?**

In most cases, the main goal of a renal replacement procedure will be to eliminate fluid and small molecular substances (potassium, creatinine, urea,  $H^+$ ); this can be done equally effectively with all three procedures. However, because haemodialysis can be performed using the lowest blood flow compared to the other two modalities (it only needs to be 2–3 times the therapeutic dose), it makes sense to use it in these cases.

If, on the other hand, the focus is on the maximally effective elimination of large-molecule substances (e.g., creatine kinase, myoglobin, interleukins), pure haemofiltration is the procedure of choice. Because a significantly higher blood flow is required here (at least five times the therapy dose), the quality of the vascular accesses is of decisive importance: haemodynamically effective constrictions in the vascular accesses can lead to strong suppressions in the vascular accesses, so that with increasing haemolysis the blood flow (and thus also the therapy dose!) may have to be limited.

Haemodiafiltration represents a compromise between the two methods: it is a good alternative if large-molecule substances are to be eliminated to a greater extent, but part of the purification performance can also be realized via the mechanism of diffusion (with the focus on small-molecule substances). The advantage over pure haemofiltration is that haemodiafiltration can be operated with lower blood flows than haemofiltration with the same therapeutic dose.

### 8.3 What Treatment Parameters?

For each renal replacement procedure, the fluid removal (fluid balance of the procedure) and the therapeutic dose (the amount of plasma fluid cleared per unit time) must be determined.

Fluid removal is highly variable and dependent on the clinical hydration status of the patient; in clinical practice, withdrawal rates between 0 and 300 ml/h are common.

The therapeutic dose, on the other hand, should always be approximately 20–30 ml/kg/h. Regardless of whether the blood is purified by filtration (CVVH), dialysis (CVVHD) or both mechanisms (CVVHDF), the fluid in the extracapillary space of the filter cartridge (filtrate, dialysate, or a mixture of both) contains all substances requiring elimination at nearly the same concentration as the plasma fluid. The therapeutic dose for all renal replacement procedures therefore corresponds to the “outflow rate”, i.e., the amount of fluid per unit of time that flows out of the extracapillary compartment of the filter cartridge.

When setting the therapy dose, it must be taken into account whether and to what extent renal replacement therapy must be interrupted, e.g., for surgical procedures, diagnostic procedures (CT, MRI) or a change of system. Because the dose guideline value of 20–30 ml/kg/h applies to continuous use, the dose set on the device should be selected correspondingly higher depending on the duration of the therapy interruptions.

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### 8.4 Which Anticoagulation?

The anticoagulation procedure of choice is regional citrate anticoagulation. It is easily controllable, does not affect the patient's in vivo coagulation and can also be used in cases of heparin intolerance.

The citrate dose is in the range of 3–4 mmol citrate per litre of blood. The exact dosage is controlled by the “postfilter”  $\text{Ca}^{++}$ ; for optimal anticoagulation, this value should be in the range of 0.25–0.35 mmol/l.

The level of subsequent  $\text{Ca}^{++}$  substitution is approximately 1.5–2.0 mmol  $\text{Ca}^{++}$  per litre of dialysate (or filtrate). The exact dosage is based on systemic  $\text{Ca}^{++}$ , which should be in the range of 1.15–1.35 mmol/l.

Citrate anticoagulation may also be used in patients with hepatic insufficiency. In this case, careful attention should be given to an increase in  $\text{Ca}^{++}$  substitution requirements as well as emerging metabolic acidosis, which indicates incomplete hepatic degradation of calcium-citrate complexes. The extent of citrate accumulation can also be objectified by the ratio of total calcium to ionized  $\text{Ca}^{++}$ ; an increase in this ratio to values above 2.5 indicates incomplete hepatic degradation of the calcium-citrate complexes. In this case, citrate anticoagulation must be discontinued and replaced by systemic anticoagulation (e.g., with heparin or argatroban).



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